INCH-POUND
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MILITARY SPECIFICATION

* CABLES, LIGHT WEIGHT, ELECTRIC, LOW SMOKE, FOR SHIPBOARD USE
GENERAL SPECIFICATION FOR

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

* 1.1 Scope. This specification covers light weight electric cables for Navy shipboard applications.

1.2 Classification. Cables covered by this specification shall be classified as watertight, non-watertight and non-watertight with circuit integrity constructions, armored and unarmored, unarmored with overall shields and further classified for power, control, communications, instrumentation and electronic applications, as specified in 1.2.1 through 1.2.3 (see 6.2.1). A part number for cable and cord shall be as specified in 6.8 and the applicable specification sheet.

1.2.1 Non-watertight, nonflexing service. Cable types for non-watertight, nonflexing service as follows:

Power:

	Unarmored	Unarmored with overall shield	Armored		
Type	DX	----	DXA	Two conductors	600 volts (V)
	FX	----	FXA	Four conductors	600 V
	TX	----	TXA	Three conductors	600 V

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, SEA 05Q42, Naval Sea Systems Command, 2531 National Center Bldg. 3, Washington, DC 20362-5160, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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Electronic, communication and instrumentation:

	Unarmored	Unarmored with overall shield	Armored		
Type	----	MXO	----	Multi-conductor,	AWG 20 600 V
	----	MXSO	----	Multi-conductor,	AWG 16 600 V
	TTX	----	TTXA	Multi-pair,	AWG 20 600 V
	TTXS	TTXSO	TTXSA	Multi-pair,	AWG 22 600 V
	----	1XMSO	----	Multi-conductor,	AWG 22 600 V
	----	2XAO	----	Multi-pair,	AWG 22 600 V
	----	2XO	----	Multi-pair,	AWG 26 600 V
	2XS	2XSO	2XSA	Multi-pair,	AWG 22 600 V
	----	2XSXO	----	Multi-pair,	AWG 26 600 V
	3XS	----	3XSA	Multi-triad,	AWG 18 600 V

1.2.2 Watertight, non-flexing service. Cable types for watertight, non-flexing service are as follows:

Electronic, communication and instrumentation:

	Unarmored	Unarmored with overall shield	Armored		
Type	----	1XSOW	----	Multi-conductor,	AWG 22 600 V
	----	2XOW	----	Multi-pair,	AWG 26 600 V
	2XSAW	2XSAOW	2XSAWA	Multi-pair,	AWG 22 600 V
	2XSW	2XSOW	2XSWA	Multi-pair,	AWG 18 600 V
	3XSW	3XSOW	3XSWA	Multi-triad,	AWG 22 600 V

1.2.3 Watertight with circuit integrity, non-flexing service. Cable types for watertight with circuit integrity, nonflexing service are as follows:

Power:

	Unarmored	Unarmored with overall shield	Armored		
Type	DXW	DXOW	DXWA	Two conductor	600 V
	FXW	FXOW	FXWA	Four conductor	600 V
	TXW	TXOW	TXWA	Three conductor	600 V
	7XW	----	7XWA	Seven conductor	600 V

Electronic, communication and instrumentation:

	Unarmored	Unarmored with overall shield	Armored		
Type	TTXW	TTXOW	TTXWA	Multi-pair, AWG 22	600 V

Control:

	Unarmored	Unarmored with overall shield	Armored		
Type	MXCW	MXCOW	MXCWA	Multi-conductor, AWG 18	600 V

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2. APPLICABLE DOCUMENTS

2.1 Government Documents.

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

TT-I-735	Isopropyl Alcohol.
PPP-V-205	Veneers, Paper Overlaid, Container Grade

MILITARY

MIL-I-631	Insulation, Electrical, Synthetic-resin Composition, Nonrigid.
MIL-H-5606	Hydraulic Fluid, Petroleum Base; Aircraft, Missile, and Ordnance.
* MIL-T-5624	Turbine Fuel, Aviation, grades JP-4, JP-5, and JP-5/JP-8 ST.
MIL-C-12000	Cable, Cord, and Wire, Electric, Packaging of.
MIL-F-16884	Fuel, Naval Distillate.
MIL-L-17331	Lubricating Oil, Steam Turbine and Gear, Moderate Service.
MIL-H-17672	Hydraulic Fluid, Petroleum, Inhibited.
MIL-L-23699	Lubricating Oil, Aircraft Turbine Engine, Synthetic Base.
MIL-C-27500	Cable, Power, Electrical and Cable Special Purpose, Electrical Shielded and Unshielded, General Specification for.
MIL-W-81044	Wire, Electric, Crosslinked Polyalkene, Crosslinked Alkene-Imide Polymer, or Polyarylene Insulated, Copper or Copper Alloy.
MIL-W-81044/12	Wire, Electric, Crosslinked Polyalkene Insulated Tinned Coated Copper, Light Weight, 600V 150°C.
MIL-W-81381	Wire, Electric, Polyimide Insulated, Copper or Copper Alloy.

(see supplement 1 for list of applicable specification sheets.)

STANDARDS

FEDERAL

FED-STD-228	Cable and Wire, Insulated; Methods of Testing.

MILITARY

MIL-STD-104	Limits for Electrical Insulation Color.
MIL-STD-454	Standard General Requirements for Electronic Equipment.

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. The issue of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

ASTM

* B 3 Soft or Annealed Copper Wire. (DoD adopted)
B 8 Concentric-Lay-Stranded Copper Conductors, Hard, Medium Hard or Soft. (DoD adopted)
* B 33 Tinned Soft or Annealed Copper Wire for Electrical Purposes. (DoD adopted)
B 193 Resistivity of Electrical Conductor Materials. (DoD adopted)
B 286 Copper Conductors for Use in Hookup wire for Electronic Equipment. (DoD adopted)
D 470 Standard Methods of Testing Crosslinked Insulation and Jackets for Wire and Cable.
D 2240 Rubber Property Durometer hardness. (DoD adopted)
D 2565 Standard Practice for Operating Xenon Arc Type (Water Cooled) Light-Exposure Apparatus Wire and Without Water for Exposure of Plastics.

(Application for copies should be addressed to ASTM, 1916 Race Street, Philadelphia, PA 19103.)

AMERICAN NATIONAL STANDARDS INSTITUTE, INC. (ANSI)

B46.1 Surface texture, Surface Roughness, Waviness and Lay. (DoD adopted)

(Application for copies should be addressed to American National Standards Institute, Inc. 1430 Broadway, New York, NY 10018.)

* INSULATED CABLE ENGINEERS ASSOCIATION (ICEA)

T-28-562 Test Method for Measurement of Hot Creep of Polymeric Insulation's.

(Application for copies should be addressed to the Insulated Cables Engineers Association, PO Box 440, South Yarmouth MA 02664)

INSTITUTE OF ELECTRICAL AND ELECTRONIC ENGINEERS (IEEE)

383 IEEE STD for Type Test of Class IE Electric Cables, Field Splices and Connections for Nuclear Power Generating Stations.

(Application for copies should be addressed to the Institute of Electrical and Electronic Engineers, Inc., 345 East 47th Street New York, NY 10017)

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Other publications continued ...

* NATIONAL ELECTRICAL MANUFACTURES ASSOCIATION (NEMA)
WC-61 Surface Transfer Impedance

(Application for copies should be addressed to the National Electrical Manufacturers Association, Suite 300, 2101 L Street, N.W. Washington, DC 20037)

NAVAL ENGINEERING STANDARDS (NES)

711 Determination of the Smoke Index of the Products of Combustion from Small Specimens of Materials, Issue 2 (1/81).
713 Determination of the Toxicity Index of the Products of Combustion from Small Specimens of Material, Issue 3 (3/85).

(Copies of Naval Engineering Standards are sponsored by the Procurement Executive, Ministry of Defense, Ship Department, Section TE112, Block G, Foxhill, Bath 5AB England.)

Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification will take precedence.

3. REQUIREMENTS

3.1 Detail specifications. The individual item requirements shall be as specified herein and in accordance with the applicable specification sheet. In the event of a conflict between the general requirements of this specification and the specification sheet, the latter shall govern.

3.2 Qualification. Cables furnished under this specification shall be products which are qualified for listing on the applicable qualified products list (QPL) at the time set for opening of bids (see 4.3 and 6.5).

3.2.1 Requalification. Changes in material and construction which have been qualified shall require the written approval of the Naval Sea Systems Command (NAVSEA). Incorporation of any changes which have not been so approve shall require requalification of the cable or groups of cables in question.

3.3 Materials. Materials used in construction of those cables requiring qualification shall have the written approval of NAVSEA. In the case of cables not requiring qualification, NAVSEA approval will not be required, and materials in these cables shall be in accordance with the requirements shown in the applicable specification sheets.

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3.3.1 Conductors. Copper conductors shall be composed of soft or annealed copper strands in accordance with ASTM B 3 or tin-coated, soft annealed copper strands in accordance with ASTM B 33. Strands shall be free from lumps, kinks, splits scraped or corroded surfaces and skin impurities.

3.3.2 Insulation. Insulation shall be as specified on the applicable specification sheet.

3.3.3 Jackets. The material used for the cable jacket shall be crosslinked polyolefin which shall have the physical and electrical properties as required by the applicable specification sheet and as specified herein. Unless otherwise specified in the applicable specification sheet, the color of the cable or cord jacket shall be black.

3.3.4 Shields. The materials and constructions for shields of insulated conductors, groups of insulated conductors and overall cable shall be as specified in the specification sheet.

* 3.3.5 Fillers. Unless otherwise specified in the applicable specification sheet, fillers for cables intended for flexing applications, only fibrous fillers shall be used. For non-flexing types, solid or fibrous filler materials shall be acceptable. Fillers for cables which are required to be watertight shall be nonfibrous or a combination of nonfibrous and fibrous.

3.3.5.1 Fibrous fillers. Fibrous fillers shall be treated for flame or moisture resistance, or a combination of both, to meet the requirements for the particular type of cable.

3.3.5.2 Asbestos fillers. The use of asbestos or asbestos form containing materials shall be prohibited throughout the construction and manufacturing of cables.

* 3.3.5.3 Nonfibrous fillers (solids). Nonfibrous fillers shall consist of elastomeric material which is removable from insulation of conductors and insulating coverings over the shields. The acceptability of the material shall be in accordance with 4.7.3 and 4.7.4.

* 3.3.6 Tapes. Unless otherwise specified (see 3.1), tapes shall be polyester, type G in accordance with MIL-I-631. Tapes applied over shield braids for singles, twisted pairs and triads, shall be sealed or set to prevent unwinding freely, but shall be easily removable for wire and/or shield termination.

3.3.7 Separators. Separators, when required, shall be of a material that is compatible with other cable materials. The compatibility of the material shall be in accordance with 4.7.3 and 4.7.4.

3.3.8 Binders. Binders shall be of a material compatible with the other cable materials. The compatibility of the material shall be in accordance with 4.7.3.

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* 3.3.9 Armor wires. The armor wires shall consist of Alclad 5056 or 5154 aluminum alloy having a diameter of 0.0126 +/- 0.0005 inch, a minimum tensile strength of 50,000 pounds per square inch (lb./in^2) and a minimum elongation (before application to the cable) of 2 percent in 10 inch length.

3.3.10 Fungus Resistance. All nonmetallic materials shall be fungus inert and shall be certified that they meet the requirements of requirement 4 of MIL-STD-454 (see 6.2.2).

* 3.3.11 Water blocking compound. Water blocking compound used in cable and shield interstices shall be compatible with all other cable materials. The compound shall be clean, nontacky to the touch and shall leave no residue on the installers hands. The compound shall be free stripping from the cable components and shield by hand and shall not require the use of chemicals or other mechanical means of removal. The compound shall not interfere with any termination technique used with finished cable, shields or components. The compatibility of the compound shall be in accordance with 4.7.3 and 4.7.4.

3.3.12 Hazardous items and toxic materials. The material shall have no adverse effect on the health of personnel when used for its intended purpose. Questions pertinent to this effect shall be referred by the contracting activity to the appropriate departmental medical service who will act as an advisor to the contracting agency. Regardless of any other requirements materials and parts containing asbestos shall not be used.

3.3.13 Materials control. All materials included in the construction of the cable shall be examined and tested to ensure conformance to this specification and applicable specification sheet. Once a cable construction has been fully qualified, no materials may be added, deleted or modified, without the written approval of NAVSEA. Any addition, deletion or modification of materials within a cable construction shall require requalification.

3.3.14 Recovered materials. Unless otherwise specified herein, all equipment, material and articles incorporated in the products covered by this specification shall be new and shall be fabricated using materials produced from recovered materials to the maximum extent practicable without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products is allowed under this specification unless otherwise specifically specified.

3.4 Design and construction.

* 3.4.1 Component wire. Finished wire shall be as specified in the applicable specification sheet, or as approved by NAVSEA

3.4.2 Separators. Separators employed directly over conductors shall be applied to give not less than 100 percent coverage to the conductors.

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3.4.3 Insulation. The insulation shall be as specified in the applicable specification sheet. Conductor insulation shall be readily removable by conventional wire stripping devices without damage to the conductor.

3.4.4 Tapes. Tape wraps over the cable core, shield and other locations shall be as specified in the applicable specification sheet. Tapes shall be applied helically with an overlap of not less than 25 percent.

3.4.5 Fillers and binders. Fillers and binders shall be used, where required, to provide firmness and roundness of completed cables and to provide watertightness when required by the specification sheet. Filler and binder material shall be compatible with all other cable components. Compatibility of components shall be determined in accordance with 4.7.3 and 4.7.4.

3.4.6 Shields. Shields, when specified in the applicable specification sheet, shall be of the push back type and shall be constructed to conform to the physical and electrical requirements specified in the applicable specification sheet. The shield shall be of a close fitting braid construction and shall be free from irregularities and discontinuities. Individual wire strands may be spliced, but not more than one carrier may be spliced at any one point. When the braided shield is cut, it shall be capable of sliding back not less than 4 inches on an exposed 24 inch length of component wire or cable with the opposite end clamped.

* 3.4.6.1 Shield insulation. Shield insulation shall consists of two polyester tapes, type G in accordance with MIL-I-631, sealed. An alternate shield insulation may consist of one tape, type G of MIL-I-631, plus an extruded jacket of a suitable translucent material, minimum average thickness of the transparent material 0.003 inch. The standard identification code shall be applied by method 2 on the inner tape.

3.4.7 Watertightness. Where watertight cable construction is specified (see 3.1), voids within the cable or cord construction shall be filled to prevent the passage of water longitudinally through the cable.

3.4.8 Braid armor. The armor shall be applied in the form of a braid. The weave of the braid shall be of either the "one over - one under" or the "two over - two under" type. The weave shall be such that the wires of a carrier are laid closely together, flat and parallel. The braid shall be applied with maximum tension possible so as to prevent loosening and creeping but not to cause broken ends. The braid shall remain snug to the cable jacket and not spring away when cut. Splices in the braid strands shall be infrequent, staggered and inconspicuous so as not to increase the nominal diameter of the cable or result in rough spots. The selection of the number of wires per carrier and the number of carriers per braid shall be such as to produce a basket weave to give a minimum of 88 percent coverage and a braid angle within the limits as specified in table I. The number of wires per carrier shall be not more than as specified in table II.

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TABLE I. Armor braid angle.

Diameter over jacket (inches)	Braid Angle	
	Minimum (degrees)	Maximum (degrees)
0.100 - 0.600	30	60
.601 - 1.000	35	60
1.001 - 1.500	40	70
1.501 - 2.000	45	70
2.001 - 3.000	50	80

TABLE II. Wires per carrier

Diameter under braid (inches)	Max. number of ends/carrier	
	One over One under	Two over Two under
0.100 - 0.400	8	5
.401 - .800	12	10
.801 - 3.000	16	10

* 3.4.8.1 Coverage. The coverage shall be as determined by the following formula:

$$\text{Percent coverage } (k) = 100\ (2F-F^2)$$

Where $F = NPd/\sin a$

$\tan a = 2 \times 3.14159\ (D+2d)P/C$

a = acute angle of braid with axis of cable or cord
d = diameter (inch) of individual braid wires
D = diameter (inch) of cable under braid
N = number of wires per carrier
C = number of carriers
P = picks per inch of cable or cord length

3.4.9 Cable jacket. A well centered jacket shall be applied over the cable core. The jacket shall conform to the applicable specification sheet and shall be removable without damage to the underlying shield or components.

3.4.10 Finished cable. Finished cable shall conform to the applicable specification sheet. Cable dimensions shall be measured as specified in 4.6.2.

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* 3.4.10.1 Cabling. The number of component wires shall be as specified in the applicable specification sheet and shall be cabled together with a left hand lay. For cables having multiple layers, the outer layer shall be left hand and the inner layer or layers may be either right hand or left hand lay. The lay length of the individual finished wires shall be not less than 8 times nor more than 16 times the diameter of the applicable layer.

3.4.10.2 Concentricity. Concentricity shall not be less than 70 percent and shall apply to both the finished wires and finished cable (see 4.7.1).

3.5 Electrical properties. Electrical properties of the completed cable shall be as specified by the specification sheet.

3.6 Physical properties. Physical properties of the completed cable and cable components shall be as specified by the specification sheet.

3.7 Identification codes and methods. Individual conductors and groups of conductors shall be separately identified in accordance with 3.7.1, 3.7.2 and 3.7.3.

3.7.1 Identification codes. Unless otherwise specified (see 6.2), the standard identification code and where applicable, twisted pair and triad identification codes shall be used.

3.7.1.1 Standard identification code. Standard identification code shall be in accordance with table III.

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TABLE III. Standard identification code.

Color, conductor or group no.	Background or base color	First tracer color	Second tracer color
1	Black	----	----
2	White	----	----
3	Red	----	----
4	Green	----	----
5	Orange	----	----
6	Blue	----	----
7	White	Black	----
8	Red	Black	----
9	Green	Black	----
10	Orange	Black	----
11	Blue	Black	----
12	Black	White	----
13	Red	White	----
14	Green	White	----
15	Blue	White	----
16	Black	Red	----
17	White	Red	----
18	Orange	Red	----
19	Blue	Red	----
20	Red	Green	----
21	Orange	Green	----
22	Black	White	Red
23	White	Black	Red
24	Red	Black	White
25	Green	Black	White
26	Orange	Black	White
27	Blue	Black	White
28	Black	Red	Green
29	White	Red	Green
30	Red	Black	Green
31	Green	Black	Orange
32	Orange	Black	Green
33	Blue	White	Orange
34	Black	White	Orange
35	White	Red	Orange
36	Orange	White	Blue
37	White	Red	Blue
38	Brown	-----	----
39	Brown	Black	----
40	Brown	White	----

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TABLE III. Standard identification code. continued

Color, conductor or group no.	Background or base color	First tracer color	Second tracer color
41	Brown	Red	----
42	Brown	Green	----
43	Brown	Orange	----
44	Brown	Blue	----
45	White	Black	Blue
46	Red	White	Blue
47	Green	Orange	Red
48	Orange	Red	Blue
49	Blue	Red	Orange
50	Black	Orange	Red
51	White	Black	Orange
52	Red	Orange	Black
53	Green	Red	Blue
54	Orange	Black	Blue
55	Blue	Black	Orange
56	Black	Orange	Green
57	White	Orange	Green
58	Red	Orange	Green
59	Green	Black	Blue
60	Orange	Green	Blue
61	Blue	Green	Orange
62	Black	Red	Blue
63	White	Orange	Blue
64	Red	Black	Blue
65	Green	Orange	Blue
66	Orange	White	Red
67	Blue	White	Red
68	Black	Green	Blue
69	White	Green	Blue
70	Red	Green	Blue
71	Green	White	Red
72	Orange	Red	Black
73	Blue	Red	Black
74	Black	Orange	Blue
75	Red	Orange	Blue
76	Green	Red	Black
77	Orange	White	Green
78	Blue	White	Green
79	Red	White	Orange
80	Green	White	Orange

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TABLE III. Standard identification code. continued

Color, conductor or group no.	Background or base color	First tracer color	Second tracer color
81	Blue	Black	Green
82	Orange	White	----
83	Green	Red	----
84	Black	Green	----
85	White	Green	----
86	Blue	Green	----
87	Black	Orange	----
88	White	Orange	----
89	Red	Orange	----
90	Green	Orange	----
91	Blue	Orange	----
92	Black	Blue	----
93	White	Blue	----
94	Red	Blue	----
95	Green	Blue	----
96	Orange	Blue	----
97	Yellow	----	----
98	Yellow	Black	----
99	Yellow	White	----
100	Yellow	Red	----
101	Yellow	Green	----
102	Yellow	Orange	----
103	Yellow	Blue	----
104	Black	Yellow	----
105	White	Yellow	----
106	Red	Yellow	----
107	Green	Yellow	----
108	Orange	Yellow	----
109	Blue	Yellow	----
110	Black	Yellow	Red
111	White	Yellow	Red
112	Green	Yellow	Red
113	Orange	Yellow	Red
114	Blue	Yellow	Red
115	Black	Yellow	White
116	Red	Yellow	White
117	Green	Yellow	White
118	Orange	Yellow	White
119	Blue	Yellow	White
120	Black	Yellow	Green

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TABLE III. Standard identification code, continued

Color, conductor or group no.	Background or base color	First tracer color	Second tracer color
121	White	Yellow	Green
122	Red	Yellow	Green
123	Orange	Yellow	Green
124	Blue	Yellow	Green
125	Black	Yellow	Blue
126	White	Yellow	Blue
127	Red	Yellow	Blue

3.7.1.2 Telephone identification code. Conductor identification code for telephone cables shall be as follows:

Color or conductor no.	Color	Color or conductor no.	Color
1	Black	7	Brown
2	White	8	Gray
3	Red	9	Yellow
4	Green	10	Purple
5	Orange	11	Tan
6	Blue	12	Pink

The pairing of conductors for forming pairs shall be as follows:

No. 1 paired with nos. 2 thru 12 for first 11 pairs
No. 2 paired with nos. 3 thru 12 for next 10 pairs
No. 3 paired with nos. 4 thru 12 for next 9 pairs
No. 4 paired with nos. 5 thru 12 for next 8 pairs
No. 5 paired with nos. 6 thru 12 for next 7 pairs
No. 6 paired with nos. 7 thru 12 for next 6 pairs
No. 7 paired with nos. 8 thru 12 for next 5 pairs
No. 8 paired with nos. 9 thru 12 for next 4 pairs
No. 9 paired with nos. 10 thru 12 for next 3 pairs
No. 10 paired with nos. 11 thru 12 for next 2 pairs
No. 11 paired with no. 12

3.7.1.3 Special identification code. Special identification code shall be in accordance with the following:

Color or conductor no.	Color	Color or conductor no.	Color
1	Black	7	Brown
2	White	8	Gray
3	Red	9	Yellow
4	Green	10	Purple
5	Orange	11	Tan
6	Blue	12	Pink

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3.7.1.4 Letter identification code. Letter identification code shall consist of the letters A, B, C and D printed in block type and with black, white, red and green ink respectively.

3.7.1.5 Twisted pair identification code. This code shall consist of the numbers in sequence running from 1 through the number corresponding to the total number of twisted pairs in the cable. Both conductors in each pair shall be numbered the same, denoting the sequence number of the pair. Distinction between the two conductors is provided by different colored insulation. Conductors of a cable with a single pair need not be numbered.

3.7.1.6 Twisted triad identification code. This code shall consist of the numbers in sequence running from 1 through the number corresponding to the total quantity of twisted triads in the cable. Three conductors shall be numbered the same, denoting the sequence number of the triad; distinction between the three conductors is provided by different colored insulation. Conductors of a cable with a single triad need not be numbered.

3.7.2 Identification methods. Unless otherwise specified (see 6.2), identification method 1 or method 3 shall be used.

* 3.7.2.1 Method 1. Identification method 1 shall be surface printing of both number and color designations. The legend shall be printed in contrasting color: preferably white ink on black or dark background or black ink on white or light background. The legend shall be repeated at intervals not exceeding 3 inches and alternate legends shall be inverted. For example 10 ORANGE BLACK BLACK ORANGE 10. Unless otherwise specified (see 3.1), the axis of the characters shall be perpendicular to the longitudinal axis of the wire. The character type shall be block or italic and shall have a height in accordance with the diameter over which it is applied as follows:

Diameter range (inch)	Height of character approximate (inch)
0.030 to 0.070	0.025
.070 to .095	.031
.095 to .115	.047
.115 to .200	.062
.190 to .250	.078
.235 to .325	.094
.330 and larger	.125

* 3.7.2.2 Method 2. Identification method 2 shall use translucent (opaque) polyester tapes (see 3.3.6) which have been printed with both the number and the color designation. The legend shall be printed with black ink and shall be repeated at intervals not exceeding 3 inches and alternate legends shall be inverted. The character type shall be block and shall be approximately 3/32 inch height.

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3.7.2.3 Method 3. Identification method 3 shall be the use of solid base colors or solid base colors with tracers as required. The base color may be either the color of the insulation or the color of a coating applied to the insulation. The tracers shall be approximately 1/32 inch wide ink stripes of the required color applied helically with 1 1/2 +/1/4 inch lay. If two tracers are required, the second shall be half the width of the first.

3.7.2.4 Method 4. Identification method 4 shall be the use of colored braids. Tracers shall consists of the required colors applied by three adjacent carriers. Where two tracers are required, they shall be applied with reverse lay.

3.7.2.5 Method 5. Identification method 5 shall be the use of the printed letter on the outermost insulating tape or the printed letter on a polyester binder tape over the insulating tapes. The letters shall be approximately 3/16 inch high and shall have been printed at intervals not exceeding 3 inches prior to the application of the tape to the conductor. If the insulating tapes are white, no printing is required on the B (white) conductor.

* 3.7.2.6 Method 6. Identification method 6 shall consist of numerals printed in ink on the conductor insulation. For conductors having a jacket directly over the insulation, the numerals may be printed in ink on the jacket, at the manufacture's option. White ink shall be used for a red or black background; black ink shall be used for a white background. Numerals may be perpendicular or parallel to the longitudinal axis of the conductor (see figure 1). Numeral width shall be proportional to conductor outside diameter (o.d.) as follows:

Diameter range (inch)	Height of character approximate (inch)
0.030 to 0.070	0.025
.071 to .095	.031
.096 to .115	.047
.115 to .200	.062
.190 to .250	.078
.235 to .325	.094
.330 and larger	.125

Numeral height shall be 2 1/2 to 3 times numeral width. Each numeric legend shall be underlined. Two digit legends which are parallel to the longitudinal axis shall have the bottom numeral underlined. Legends shall be alternately inverted and shall be repeated at intervals not greater than 1 1/2 inches.

3.7.3 Manufacturer's identification tape. Unless otherwise indicated on the specification sheet, cables shall contain a continuous, thin, moisture resistant marker tape, not less than 1/10 inch wide. The marker tape shall be placed directly under the cable or cord binder tape or jacket. The tape shall be printed to show the following at intervals not greater than 1 foot: Name and location of manufacture; year of manufacture; specification number (MIL-C-24640); and progressive serial number. The serial number is not necessarily a footage marker. A serial

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number shall not be repeated by a manufacture in any one year for any one type and size of cable.

* 3.7.3.1 Marker threads. The use of marker threads in lieu of the manufacturer's identification tape, as specified in 3.7.3, in small cables cords shall have the written approval of NAVSEA. If approved. marker threads shall be in accordance with those assigned by Underwriter's Laboratory (UL) Incorporated to indicate the manufacture of the cable or cord. When UL marker threads are used, yearly marker threads shall also be used to indicate the year of manufacture, as follows:

Color codes

Year	Background or base color	Color of tracer
1994	Orange	Green
1995	Orange	Brown
1996	Orange	Black
1997	Orange	Pink
1998	White	Yellow
1999	White	Slate
2000	White	Green
2001	White	Brown
2002	White	Black

* 3.7.4 Cable or surface marking. Unless otherwise specified (see 3.1), ink marking shall be used for overall cable jacket surface marking. The legend shall be printed in contrasting color: preferably white ink on black or dark background or black ink on white or light background. The legend shall consist of the manufacturer's name, the cable type and size designation, specification sheet number, the year of manufacture and jacket type. The legend shall be repeated at intervals not exceeding 1 foot and the year of manufacture need not be in line with the balance of the legend. For cable constructions on which armor is to be applied, the unarmored or armored cable designation shall be applied on the cable jacket. For armored constructions, Permanence of Printing (Jacket), 4.7.12, requirements are not applicable.

For example: "Manufacturer's name - DX-3-MIL-C-24640/1-1991 XLPOLYO

The character type may be either block or italic and shall have a minimum height in accordance with the diameter over which it is applied as follows:

Diameter range (inch)	Height of character approximate (inch)
0.125 to 0.200	0.047
.200 to .285	.063
.285 to .350	.078
.350 to .500	.094
.500 and larger	.125

Jacket material name shall be abbreviated as follows: XLPOLYO

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility of inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Inspection system. The contractor shall provide and maintain an inspection system acceptable to the Government for supplies and services covered by this specification. The inspection system shall be in accordance with the ordering documents (see 6.2.2).

4.2 Classification of inspection. The examination and testing of all cable and cord covered by this specification shall be classified as follows:

(a) Qualification inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

4.3 Qualification inspection. Qualification inspection shall consist of the examination and test specified in tables IV and VI and in accordance with the applicable specification sheet.

TABLE IV. Qualification Inspection.

Examination or Test	Requirement para.	Test method para.
Cable aging and compatibility	Specification Sht	4.7.3
Acid gas equivalent	Specification Sht.	4.7.16
Halogen content	Specification Sht.	4.7.17
Flame propagation	Specification Sht.	4.7.8
Immersion	Specification Sht.	4.7.19
Smoke index	Specification Sht.	4.7.18
Toxicity index	Specification Sht.	4.7.20
Durometer	Specification Sht.	4.7.7
Weathering	Specification Sht.	4.7.21
Fungus resistance	3.3.10	----
Abrasion resistance	Specification Sht.	4.7.1
Gas Flame (3 hour)	Specification Sht.	4.7.9
Finished wire	Specification Sht.	Specification Sht.

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* 4.3.1 Qualification specimens. A manufacturer seeking qualified products listing (QPL), shall manufacture and be responsible for testing a cable type and size required for group qualification in accordance with table V. When a manufacturer desires qualification of an individual type, size or group, the selection of test specimens and scope of qualification shall be subject to review by NAVSEA. For slash sheets (see 3.1) having more than one type of insulation compound, each insulation type will require qualification by NAVSEA. A sufficient length of any test specimen shall be manufactured at one time. Qualification will apply for armored, unarmored and unarmored with overall shielded cables.

* 4.3.1.1 Authorized testing of qualification samples outlined in Table V or testing of selected samples to incorporate alternate materials in the originally approved Details of Construction Sheets and approval of low smoke electrical cables based on satisfactory test results for the representative samples shall not in any way be construed as meaning that only the type/size/variation of the representative sample(s) has to meet the specification requirements. All types/sizes/variations comprising the group(s) represented by the test sample(s) must meet all specification requirements.

* 4.3.2 Qualification retention. In order to maintain qualification, each cable manufacturer shall annually perform the following tests in accordance with the applicable test paragraph, (see Table IV), on all materials for which a requirement is specified.

- Halogen content
- Toxicity
- Acid gas
- Smoke testing
- Immersion

TABLE V. Qualification test samples and groups.

Qualification test samples			
Group No.	Additional Cable types and sizes for flame propagation test only	All tests specified in the applicable specification sheet type and size	Types comprising the Group 1/
1	None	DX-3	DX, TX AND FX
2	DWX-3	MXCOW-10	DXW, TXW, FXW, DXOW, TXOW, FXOW, MXCOW, 7XW AND MXCW
3	None	2XSAOW-10	1XSOW, 2XSAW, 2XSW AND 3XSW
4	None	2XOW-6 OR 2XOW-42	2XOW
5	TTX-15	MXO-10	MXO, MXSO, TTX, TTXS, 2XAO, 1XMSO, 2XS, 3XS, 2XO AND 2XSXO
6	TTXW-15 or TTXW-20	TTXOW-15 OR TTXOW-20	TTXW

1/ Includes all sizes and variations of the cable type.

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* 4.4 Quality conformance inspection. Quality conformance inspection shall be performed on all completed cable in accordance with the procedure specified herein. This inspection shall consist of basic electrical test plus groups A, B, C and D examination and tests specified in table VI and in accordance with the specification sheets.

* TABLE VI. Quality conformance inspection.

Examination or test	Requirement paragraph	Test method paragraph
Basic electrical:		
Conductor resistance	Specification sheet	4.8.4
Voltage withstand	Specification sheet	4.8.8
Insulation resistance	Specification sheet	4.8.5
Conductor and shield continuity	Specification sheet	4.8.11
Jacket flaws	Specification sheet	4.8.9
Group A:		
Visual and dimensional	3.1, 3.2, 3.3, 3.4, 3.5, 3.6, 3.7 and 3.8	4.6
Capacitance	Specification sheet	4.8.2
Characteristic impedance	Specification sheet	4.8.3
Watertightness	Specification sheet	4.7.15
Weight	Specification sheet	4.7.24
Group B:		
Cold bending	Specification sheet	4.7.5
Cross link proof	Specification sheet	4.7.25
Drip	Specification sheet	4.7.6
Tear strength	Specification sheet	4.7.14
Physicals (unaged)	Specification sheet	4.7.13 and 4.7.22
Attenuation	Specification sheet	4.8.1
Gas Flame (1 hour)	Specification sheet	4.7.9
Group C:		
Physicals (aged)	Specification sheet	4.7.13 and 4.7.22
Permanence of printing (conductor insulation)	Specification sheet	4.7.11
Permanence of printing (jacket)	Specification sheet	4.7.12
Armor	Specification sheet	4.7.2
Heat distortion	Specification sheet	4.7.10
Cable filler removability	Specification sheet	4.7.4
Shrinkage	Specification sheet	4.7.23
Mutual inductance	Specification sheet	4.8.6
Surface transfer impedance	Specification sheet	4.8.11
Pulse response time	Specification sheet	4.8.7
Group D:		
Flame propagation	Specification sheet	

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* 4.4.1 Inspection lot. An inspection lot shall consist of the total number of units of product of any one type, size and construction manufactured under essentially the same conditions and offered for delivery at one time.

* 4.4.2 Basic electrical tests. Basic electrical tests shall be in accordance with the applicable specification sheet and shall be performed on each length of completed cable. For electrical test purposes, length of completed cable shall be as defined in paragraph 6.7.1.1.

* 4.4.3 Sampling procedure. The required number of samples (see 6.7.2) for groups A, B, C and D examination and tests shall be selected at random from the inspection lot. Nonconforming starting and finishing ends of cable shall be removed by the manufacturer prior to selecting samples.

4.4.3.1 Sampling for group A examination and tests. Samples for group A examination and test shall be selected from each lot as specified in table VII.

* TABLE VII. Sampling for group A examination and tests.

Units of Product in lot	Number of Samples
1	1
2 thru 10	2
11 thru 30	3
31 thru 90	7
91 thru 210	15
211 and over	25

4.4.3.2 Sampling for group B tests. Samples for group B tests shall be selected from each as specified in table VIII.

TABLE VIII. Sampling for group B tests.

Units of product in lot	Number of samples
8 and under	1
9 thru 30	2
31 thru 90	3
91 thru 210	4
211 and over	5

4.4.3.3 Sampling for group C tests. Samples for group C tests shall be selected as specified in table IX.

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TABLE IX. Sampling for group C tests.

Two month's production (units of product)	Number of samples
8 and under	1
9 thru 30	2
31 thru 90	3
91 thru 210	4
211 and over	5

* 4.4.3.4 Sampling for group D tests. Samples for group D test shall be selected by testing the first production run of each type and size of cable and once for each 75,000 feet there after.

* 4.4.3.5 Sampling, accept/reject criteria. Material found unacceptable shall be resubmitted for reinspection only after all materials are re-examined and or re-tested and all defective materials are removed or defects corrected. Material resubmitted shall be verified 100% for the failing characteristic. Material found acceptable during rescreening inspection/testing can be re-identified and released as a separate lot.

4.4.4 Group A examination and tests. Group A examination and tests as required by the specification sheet shall be performed on samples selected in accordance with 4.4.3.1.

4.4.5 Group B tests. Group B tests as required by the specification sheet shall be performed on samples selected in accordance with 4.4.3.2.

4.4.6 Group C tests. Group C tests as required by the specification sheet shall be performed on specimens taken from samples selected in accordance with 4.4.3.3.

* 4.4.7 Group D test. Group D tests as required by the specification sheet, shall be performed on specimens taken from samples selected in accordance with paragraph 4.4.3.4.

* 4.4.8 Certified test reports. The contractor shall prepare test reports in accordance with the data ordering document (see 6.2.2). The test report shall contain the following information:

(a) Manufacturer's QPL number and date or serial number and date of NAVSEA letter of approval.
(b) A statement to the effect that the product was constructed from materials listed on the manufacturer's approved details of construction sheet.
(c) A statement that the product meets all of the requirements of this specification.
(d) Results of all quality conformance tests showing actual values obtained.
(e) Year and month cable was manufactured.
(f) Serial numbers of the marker tape taken from each end of each length of cable.

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(g) Customer's name and contract or order number.

* (h) A statement that the product contains no metallic mercury or mercury compounds and are free from mercury contamination.

* (i) The serial number of all NAVSEA letters approving deviations from the approved specification slash sheet.

* 4.5 Test conditions. Unless otherwise specified (see 3.1), the examination and tests specified in tables IV and VI shall be made at standard ambient conditions as follows:

(a) Temperature: 23 +/- 20 °C.
(b) Humidity: 50 +/- 40 % RH

4.6 Visual and dimensional examination. Cables shall be visually examined to verify that the materials, design, construction, physical dimensions, marking and workmanship are as specified in 3.1, 3.2, 3.3, 3.4, 3.5, 3.6, 3.7 and 3.8. Concentricity, wall thickness and dimensional examinations shall be made as follows:

4.6.1 Concentricity and wall thickness. The concentricity of the finished wire and the finished cable jacket shall be determined in accordance with 4.6.1.1, as applicable. Wall thickness measurements shall be made on cross sections of the finished wire or cable jacket under suitable magnification. The wall thickness shall be the shortest distance between the outer rim of the finished wire insulation or cable jacket and the outer rim of the underlying conductor or shield strand.

* 4.6.1.1 Component wire and cable jacket. The concentricity of the finished wire and cable jacket shall be determined by first locating and recording the minimum wall thickness measured on a cross section of the finished wire and cable jacket. From this point on the outer rim of the finished wire and cable jacket (at which the minimum wall thickness was measured), three more reference points (90 degrees apart) on the outside rim of the finished wire and cable jacket shall be established. At each of these three reference points, the nearest member of the conductor shall be selected, and the minimum wall thickness be measured. The average of the four readings shall be the average wall thickness. The percent concentricity shall be defined 100 times the ratio of the minimum wall thickness to the average wall thickness.

* 4.6.2 Dimensions. Measurements shall be made on not less than a 12 inch length of cable taken from the end of the sample unit. Inner components shall be made accessible by stripping and removing the outer components so as not to nick, cut, cold work or otherwise damage the component to be measured. Four points for measurement shall be located 3 to 4 inches apart along the finished wire specimen or finished cable specimen lengths, as applicable. Measurements shall be made at each point at two perpendicular planes or as required to assure the minimum and maximum reading is attained at each point. A total of eight (8) measurements shall be performed on each specimen. The minimum, maximum and average value shall be recorded, as applicable. Measurements shall be made with a micrometer, caliper or equal. A PI-tape may be used for measurements of diameters greater than 0.250 inches.

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4.6.3 Conductor strand inspection. Before individual strands are assembled into a conductor, certification shall be provided to indicate conformance to the applicable ASTM standard reference in 3.3.1. (see 6.2.2).

4.7 Test methods (physical).

4.7.1 Abrasion resistance. Abrasion resistance of cable jacket material shall be determined as specified in 4.7.1.1 through 4.7.1.4.

* 4.7.1.1 Specimens. One cable jacket test specimen shall be prepared by extruding material onto a 16 AWG (19/29) conductor. The material shall be manufactured using process conditions as close as possible to those used to produce the overall cable jacket and shall have an overall cable diameter of 0.119 + 0.001, - 0.000 inch.

4.7.1.2 Special apparatus. Apparatus shall include the following:

(a) Abrading machine - A cylinder, which incorporates 2 abrading elements on its surface, which is motor driven to rotate about a horizontal axis at 17 plus 3, minus 2 revolutions per minute (rpm) and over which the specimen is draped, as shown in figures 2 and 3 and as described below.

(1) Each abrading element shall consist of a 5/16 inch square, (in^2) high speed tool bit (Cleveland Twist Drill Company number 855 or equivalent) which has been ground on two adjacent, longitudinal sides to produce a single, sharp 90 degree longitudinal edge free of nicks. A medium grade grinding wheel (Norton Company number 39C6018VK or equivalent) shall be used for this grinding. Abrading elements shall be reground as required; in no case shall an abrading element be used in excess of 5 hours without regrinding. Whenever the perpendicular distance between either pair of opposite longitudinal sides of an abrading element become less than 0.3085 inch (as by repeated regrinding) then that abrading element shall be discarded and replaced.

(2) The cylinder shall be 8 ± 1/8 inches in diameter and shall be rigidly fabricated from metal.

(3) Each of the 2 abrading elements shall be maintained in electrical contact with the metal cylinder by securing each in a notch cut into the cylinder surface, parallel to the cylinder axis; the 2 notches required shall be spaced 180 ± 2 degrees apart around the circumference of the cylinder. These notches shall be cut such that the sharpened 90 degree longitudinal edge of each element shall be facing outward from the cylinder surface and such that the midpoint of a straight line drawn between the midpoint of the 2 longitudinal edges adjacent to the sharpened edge is tangential + or - 0.003 inch to the cylinder surface.

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(4) An automatic counter shall be provided to totalize the number of times the test specimen is scraped by the abrading elements during the test.
(5) A 1 pound ± 0.5 ounce weight shall be provided for applying tension to the specimen.

(b) Short circuit monitor - A voltage source of not less than 12 V (either direct current (DC) or root mean square (RMS) alternating current (AC)) which can be applied between the specimen conductor and both abrading elements of the abrading machine as shown in figure 2. (The connection to the abrasion elements may be provided by a wiping contact on the cylinder.) A means (such as an electrical relay) shall be provided whereby an electrical connection between either abrading element and the specimen conductor shall automatically stop the rotation of the cylinder on the abrading machine (as by removing motor power).

4.7.1.3 Procedure. Specimens shall be wiped with a clean dry cloth to remove any lubricant or dirt. Each specimen shall then be tested (either simultaneously or at one time) as follows. Each specimen shall be hung circumferentially over the cylinder of the abrading machine such that each specimen shall form an arc of not less than 170 degrees around the cylinder for the remainder of the test. One end of each specimen shall then be secured to a fixed surface. The abrading machine weight shall be attached to the opposite specimen end such that both this specimen end and the weight be freely suspended. The automatic counter shall be set initially to zero, the short circuit monitor shall be applied between both abrading elements on the abrading machine and the conductor of each specimen. The cylinder motor shall then be turned on, allowing the cylinder to rotate beneath each specimen, from secured end to weighted end.

* 4.7.1.4 Observation. Specimen failure shall be construed if electrical contact occurs between either abrading element and any specimen conductor (as evidenced by cessation of cylinder rotation) prior to completing 150 abrasive scrapes (75 cycles).

* 4.7.2 Armor. Cable armor performance shall be determined as specified in 4.7.2.1 through 4.7.2.4.

4.7.2.1 Specimens. Each specimen shall consist of a single piece of armor wire, of not less than 6 inches in length, which has been removed from a spool of armor wire prior to cable armoring. Each specimen shall be removed from a different spool. Ten specimens shall be required.

4.7.2.2 Special apparatus. Apparatus shall include the following:

(a) Mandrel - A 1/4 ± 1/64 inch diameter steel cylinder, of length and construction for the springiness test as specified in 4.7.2.3.
(b) Scale - A spring or other type scale, the measurement error of which shall not exceed + or - 5 percent for use in the springiness and toughness tests specified in 4.7.2.3.

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(c) Special vise - A special vise, with 2 parallel edges of 0.030 ± 0.001 inch radius between which a specimen may be clamped to perform the toughness test as specified in 4.7.2.3.

4.7.2.3 Procedure. The two following test shall be performed using not fewer than 5 specimens. The specimen shall be tested one at a time for each test. The same specimen shall not be used for more than one test.

(a) Springiness - One specimen shall be selected and secured by one end to the mandrel such that it is both in contact with the curved surface of the mandrel and perpendicular to the mandrel axis. The opposite end of the specimen shall be attached to the scale, the specimen shall be wrapped circumferentially around the mandrel, by handling the scale only, for not fewer than 4 complete turns, while maintaining a specimen tension of not less than 100 nor more than 125 grams. Care should be taken to pull the scale in such a direction that it accurately indicates the specimen tension. The specimen shall then be released from the scale and allowed to uncoil. Any specimen which is kinked at any place other than the attachment points at either end shall be considered an invalid specimen, the test shall be repeated using a different specimen. This test shall be successfully repeated 4 more times (five times total), using a different specimen each time.

(b) Toughness - One specimen shall be selected and secured by one end between the rounded, horizontal and parallel vise edges, such that not less than 2 inches of the specimen extends vertically above the vise. The scale shall then be attached to the free end of the specimen and a specimen tension of 100 ± 10 grams shall be established by handling the scale only. The specimen shall be bent 90 + 15, -0 degrees alternately in both directions perpendicular to the vise edges, while maintaining the specified vise tension. Not fewer then 10 90 degree bends shall be made unless the specimen breaks sooner. The bending rate shall not exceed fifty 90 degree bends per minute. This test shall be repeated four (4) more times (five times total), using different specimens each time.

4.7.2.4 Observation. Either of the following shall constitute specimen failure:

(a) Springiness - The o.d. of the coil formed by any of the specimens around the mandrel, as measured at a middle portion of the specimen, exceeds 3/8 inch.
(b) Toughness - Any specimen withstands fewer than ten 90 degree bends without breaking.

4.7.3 Cable aging and compatibility (125°C). This test shall be to detect any significant degradation due to component incompatibility or prolonged over heating of completed cable containing thermosetting or tape insulation. This test permits either of two methods (the current overload method or the hot air oven method) to be used for heat aging.

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4.7.3.1 Specimen. Specimens for cable aging and compatibility tests shall be as follows:

(a) For the current overload method, the specimen shall consist of a 40 ± 4 foot length of completed cable.
(b) For the hot air method, 2 specimens of completed cable shall be required. Specimen number 1 shall have a length of 30 ± 3 feet. Specimen number 2 shall have a length of not less than the sum of 12 inches plus 20 times the maximum specified overall diameter of the cable (see 3.1). Both specimens shall be removed from the same length of cable.

4.7.3.2 Special Apparatus. Apparatus shall include the following:

(a) The oven (required for either heat aging method) shall support the requirements as specified in 4.7.3.3, as applicable. The oven air temperature shall be measured in the immediate vicinity of the specimen.
(b) The mandrel (required for either heat aging method) shall be rigid, smooth cylinder or partial cylinder, with a continuous curved surface of not less than 180 degrees, of suitable length and construction for the specimen bending as specified in 4.7.3.5. The mandrel diameter shall be approximately 12 times the specified maximum overall cable or cord diameter (see 3.1).
(c) Voltage withstand test apparatus (required for either heat aging method) shall be as specified in 4.8.8.3
(d) Insulation resistance test apparatus (required for either heat aging method) shall be s specified in 4.8.5.2.
(e) Thermocouple device (required for the current overload method if submethod 4.7.3.3(a)(1) is used) and an associated temperature indicating device shall be provided for the use and temperatures specified in 4.7.3.3(a). Temperatures shall be measured with an error of not more than + or - 1°C.
(f) Current source (required for current overload method only) shall be an adjustable source of either dc or 60 Hz single phase ac, for maintaining specimen conductors at an elevated temperature as specified in 4.7.3.3(a). A means shall be provided for measuring the current produced by this source with an error of not more than + or - 5 %.
(g) Resistance meter (required for the current overload method if submethod 4.7.3.3(a) is used) shall be provided for measuring the resistance of a single specimen conductor at room temperature and at elevated temperatures, as specified in 4.7.3.3(a). The meter shall exhibit a measurement error of not more than + or - 1 %.

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4.7.3.3 Procedure. Select and perform either of the two heat aging methods, the current overload method or the hot air method, specified in (a) or (b) herein respectively. The oven air temperature (in both methods), conductor temperature and current (in the current overload method only) shall be recorded at intervals of not more than 15 minutes during the first hour after attaining the heat aging temperature, at intervals of not more than 1 hour for the following 5 hours and not less than twice daily thereafter for the duration of the heat aging. The interval between the consecutive temperature recordings shall not exceed 17 hours at any time during heat aging.

(A) Prior to heat aging, the current overload method specimen shall be subjected to the insulation resistance and voltage withstand test in accordance with 4.8.5 and 4.8.8, respectively. The voltage withstand test shall use the specified voltages (see 3.1). Following these, specimen conductor ends shall be electrically interconnected such that a single series electrical circuit is formed in the conductors. Provision shall then be made to measure specimen conductor temperature, using either of the following submethods.

(1) The smallest practicable "V" shaped cut shall be made along the specimen at a point approximately midway between the specimen ends, the thermocouple shall be placed beneath the resulting triangular tab. The thermocouple shall be positioned such that it is in firm contact with any single specimen conductor layer. (For specimens with only a single layer of conductors, any specimen conductor may be used.) The thermocouple shall be used with its associated temperature indicating device to measure specimen conductor temperature during heat aging.

(2) The resistance meter shall be connected to measure the resistance between the ends of any single specimen conductor in the innermost conductor layer. (For specimens with only a single layer of conductors, any specimen conductor may be used.) The meter leads shall be positioned on the conductor extremities such that all conductor interconnections are excluded from resistance measurements. The conductor temperature shall then be determined by using the following formula:

$$\text{Conductor temperature, in } ^{o}C = (234.5 + t)\ (R/r) - 234.5$$

Where: R = Measured conductor resistance, in ohms during heat aging.
t = Measured room temperature, in ^{o}C, to which the specimen is exposed prior to heat aging. Temperature t shall be measured in the immediate vicinity of the specimen; measurement accuracy shall be within $\pm$ 0.5^{o} C.
r = Measured conductor resistance, in ohms, when the specimen is exposed to room temperature t. t shall be held constant, within $\pm$ 0.5 ^{o} C for a period of not less than 2 hours prior to this measurement.

The specimen shall next be placed into the oven such that a length of not less than the sum of 12 inches plus 20

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times the maximum specified overall cable or cord diameter (see 3.1) remains straight and horizontal and such that the remainder of the specimen is formed into the largest practicable horizontal coil (to permit free circulation of convection air currents). The current source shall be connected between the two end conductors of the single series circuit within the specimen. The specimen shall then be subjected to heat aging; the air temperature within the oven shall be raised to 50 +/- 3° C and the current source adjusted to produce a conductor temperature of 125 +/- 5° C. These temperatures shall be maintained for a continuous period of not less than 400 hours. The current source shall not be shut off during heat aging, except for momentary shut off for taking resistance measurements when submethod (a) (2) herein is used. Following heat aging, the specimen shall be allowed to cool throughout to room temperature. The conductor temperature shall then be recorded. A recording which differs from room temperature by more than 5° C shall be considered to indicate failure of the conductor measurement means. If any such failure should occur, the heat aging test shall be considered to be invalid and shall be repeated using a different specimen removed from the same cable or cord. (A possible advance indication of such a failure and the need to retest, may be the need to make prominent adjustments of conductor current in order to maintain indicated conductor temperature during heat aging.) Following heat aging, the specimen shall be subjected a second time to the insulation resistance and voltage withstand tests as specified in 4.8.5 and 4.8.8, respectively. A sample, of length not less than 12 inches plus 20 times the maximum specified overall cable or cord diameter (see 3.1), shall the be removed from that portion of the specimen which had been kept straight during the heat aging. This sample shall be subjected to the bending procedure as specified in 4.7.7.5. A portion of the remainder of the specimen shall be subjected to the cable filler removability test as specified in 4.7.8.

(b) Prior to heat aging, hot air oven method specimen 1 shall be subjected to the insulation resistance and voltage withstand tests as specified in 4.8.5 and 4.8.8, respectively. The voltage withstand test shall use the voltages specified (see 3.1). Both specimens shall then be placed within the oven, specimen number 1 shall be formed into the largest practicable horizontal coil within the oven and specimen number 2 shall be positioned such that it is both straight and horizontal. Both specimens shall be simultaneously subjected to heat aging. The air temperature within the oven shall be raised to 125 +/- 3° C. This temperature shall be maintained for a continuous period of not less than 400 hours. Following heat aging both specimens shall be allowed to cool to room temperature. Specimen 1 shall then be subjected a second time to the insulation resistance and voltage withstand tests as specified in 4.8.5 and 4.8.8, respectively. A portion of specimen 1 shall next be subjected to the cable filler removability test as specified in 4.7.8.

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Specimen 2 shall be subjected to the bending procedure as specified in 4.7.3.5.

4.7.3.4 Observation. Any of the following shall constitute specimen failure:

(a) The falling away of any material from either end of any specimen during heat aging.
(b) Jacket sagging on any unarmored specimen.
(c) Jacket exudation through the armor of any armored specimen. (Jacket bulging, without actual flow or cutting, is not considered to be exudation.)
(d) Specimen (specimen 1 if the hot air oven method is used) fails either of the insulation resistance tests (see 4.8.5).
(e) Specimen (specimen 1 if the hot air oven method is used) fails either of the voltage withstand tests (see 4.8.8).
(f) Specimen (specimen 1 if the hot air oven method is used) fails the cable filler removability tests (see 4.7.8).

4.7.3.5 Further procedure. The mandrel shall be secured to a fixed surface, one end of the specimen shall be secured such that it is both in contact with the curved surface of the mandrel and perpendicular to the mandrel center of the curvature axis. The opposite end of the specimen shall be bent circumferentially around the curved mandrel surface through an arc of not less than 170 degrees. Bending shall be accomplished at the rate of approximately 20 degrees per second. The specimen shall then be secured to maintain its bent shape (as with a piece of twine, stretched taut and tied between each end of the specimen) and shall be removed from the mandrel. Not less than two thirds of the bent portion of the specimen shall then be dissected, each of its constituent components shall be visually inspected for deterioration.

4.7.3.6 Further observation. Specimen failure shall be construed if the specimen, upon dissection and inspection, exhibits visible deterioration of such a nature or extent as to impair the performance of the cable or cord in service. Deterioration sufficient to constitute specimen failure shall include, but not limited to, distortion or cracking of any constituent component, hardening of filler material or any discoloration indicating material incompatibility.

* 4.7.4 Cable sealant removability. This test shall determine whether or not sealant materials can be readily removed from the constituent components of cable. This test shall not apply to sealant material used to fill voids between either conductor strands or braid strands.

4.7.4.1 Specimen. The specimen shall be a 2 foot length of completed cable.

4.7.4.2 Procedure. Overall specimen components (such as armors, jackets, overall binders or shields) shall be removed from one end of the specimen for a distance of not less than 10 inches, thereby exposing the cable core. Using fingers only, sealant material shall then be removed from all exposed insulated conductors or from all groupings of insulated conductors which have a common covering (such as insulated conductor

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pairs or triads, which have an overall shield or binder) for a distance of not less than 8 inches. Tools, cloths or solvents shall not be used to assist with sealant removal. The common covering and sealant material shall then be removed from not less than one grouping of insulated conductors, on specimens incorporating such groupings, for distance of not less than 5 inches. Fingers only shall then be used for sealant removal. If occasional particles of sealant material remain on any constituent component of the cable core, then the removal of these particles shall be attempted by means of light brushing with fingers or with a dry cloth.

* 4.7.4.3 Observation. Any of the following shall constitute specimen failure:

(a) Sealant material is nonflexible.
(b) Sealant material which adheres to the finger or to any component of the cable core. The adherence of occasional small particles of sealant material to the cable core is acceptable unless there are small particles which cannot be removed from the core components by light brushing with fingers or a dry cloth.
(c) Sealant material which leaves a residue on cable core components.
(d) Sealant material can not be removed in less than 5 minutes.

4.7.5 Cold bending cable. This test shall determine the ability of completed cable, which is not intended to be flexed during use, to withstand bending at reduced temperature, such as might be encountered during shipboard construction or repair.

4.7.5.1 Specimen. The specimen shall consist of a piece of completed cable which shall have a length of not less than the sum of 12 inches plus 1.6 times the specified mandrel diameter (see 3.1).

4.7.5.2 Special apparatus. Apparatus shall include the following:

(a) Refrigeration chamber shall support the requirements as specified in 4.7.5.3. The chamber air temperature shall be measured in the immediate vicinity of the specimen.
(b) Mandrel shall be rigid, smooth cylinder or partial cylinder, with a continuous, curved surface of not less than 180 degrees, of suitable length and construction for the specimen bending of 4.7.5.3. Unless otherwise specified (see 3.1) the mandrel diameter shall be 12 times the diameter of the cable. The mandrel shall be provided with a clip for affixing one specimen end (see 4.7.5.3).

4.7.5.3 Procedure. The specimen shall be straightened. if it does not remain straight of itself, it shall be held straight by securing it (as with twine to a straight wooden bar. The straightened specimen (including wooden bar if applicable) shall be placed within the refrigeration chamber and subjected to an air temperature of minus 20 $^{+}/_{-}$ 2° C (unless otherwise specified, see 3.1) for a continuous period of not less than 6 hours. Prior to any change of chamber temperature, the specimen shall be removed from the chamber (following detachment from the wooden bar, if applicable) and one specimen end shall be inserted into a clip which has been affixed to (or adjacent to) the mandrel, such that the specimen is both in contact with the curved surface of the mandrel

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and also perpendicular to the mandrel center of curvature axis. The opposite end of the specimen shall then be bent circumferentially around the curved mandrel surface through a arc of not less than 170 degrees. This bending shall be accomplished at a rate of not less than 20 degrees per second. The period of time between the removal from the refrigeration chamber and the completion of specimen bending shall be not greater than 40 seconds. Care shall be taken to avoid heating the specimen, as by excessive handling, during removal from the refrigeration chamber and during bending. Following bending, the specimen shall be secured to maintain its bent shape (as with a piece of twine stretch taut and tied between each end of the specimen) and shall then be removed from the mandrel. Not less than two thirds of the bent portion of the specimen shall then be dissected and each of its constituent components visually inspected for deterioration.

4.7.5.4 Observation. Specimen failure shall be construed if the specimen either during bending or upon dissection and inspection, exhibits distortion or cracking of any specimen component, or exhibits any other visible deterioration of such a nature or extent as to impair the performance of the cable in service.

4.7.6 Drip. This test shall determine whether or not fillers will exude and drip from the cable ends when cables containing these fillers are subjected to high temperatures. Such sealant dripping may produce deleterious effects, such as the coating or bridging of underlying electrical contacts.

4.7.6.1 Specimen. The specimen shall consist of an 18 +/- 1/2 inch length of completed cable.

4.7.6.2 Special apparatus. Apparatus shall include a heating chamber for supporting the requirements as specified in 4.7.6.3. The chamber air temperature shall be measured in the immediate vicinity of the specimen.

* 4.7.6.3 Procedure. The specimen shall be straightened and shall be suspended from one end within the heating chamber, such that the specimen hangs freely downward. The temperature within the chamber shall then be raised to 95°C ± 1° C. This chamber temperature shall be maintained for a continuous period of not less than 18 hours. Provision shall be made to catch any material which may fall away from the lower specimen end during this test.

4.7.6.4 Observation. Specimen failure shall be construed if any material falls away from the lower specimen end during the test.

4.7.7 Durometer hardness. This test shall be to determine whether or not jacketing material exhibits sufficient hardness (see 4.7.7.4) for use in cable construction.

4.7.7.1 Specimen. The specimen shall consist of a block of material of the dimensions specified in ASTM D 2240, which shall have been cured in effectively the same manner as when used in cable manufacture.

4.7.7.2 Special apparatus. Apparatus shall include a type A durometer, which shall be in accordance with ASTM D 2240.

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4.7.7.3 Procedure. The specimen shall be tested in accordance with ASTM D 2240.

4.7.7.4 Observation. Specimen failure shall be construed if the specimen exhibits a durometer hardness other than that specified (see 3.1).

4.7.8 Flame propagation. This test shall to be to determine the relative ability of grouped cable to resist the propagation of fire along its length.

* 4.7.8.1 Specimen. Each specimen shall consist of a piece of completed cable of not less than 8 feet in length. The total number of specimens required shall be as specified in the vertical flame test as specified in IEEE 383. The specimens mounted on the cable tray shall be conditioned at a minimum temperature of 18°C.

* 4.7.8.2 Special apparatus. Apparatus shall include all equipment specified in the vertical tray flame test in accordance with IEEE 383 with the addition of the following.

a. Flowmeters. A flowmeter shall be located in the propane line and in the air line feeding the burner to measure the flow rate of these gases during the test. Measurements shall be accurate to within 3 %. A mass flow controller with a output that can be recorded shall be permitted for use.
b. Cable ties. Copper or steel wire not larger than 14 AWG.
c. Air. The air to the burner shall be compressed air, either bottled or supplied through a compressed air system. The air supplied shall be filtered, when necessary, to eliminate contaminants that could effect the test results.
d. Propane. The propane gas supplied to the burner shall be equivalent to propane HD-5 as specified in ANSI/ASTM D1835-1987 [1] or CAN/CSGB-3.14-M1988 (grade 1) [3]. It shall have a heat equivalent of approximately 46.4 MJ/kg (20 k Btu/lb.) (85.0 MJ/m^3 at 20° C, 101kPa).

* 4.7.8.3 Procedure. The specimen shall be tested as specified in the vertical tray flame test in accordance with IEEE 383 except as modified below;

a. The incoming air to the chamber shall be within the following limits:
   Temperature 23°C ± 20°C
   Humidity 50% ± 40% RH
b. Air movement: The maximum air movement within the enclosure, with only the intake and exhaust openings open, the exhaust fan on, if applicable, and the burner off shall not exceed 1 m/s (3.3 ft/sec.), as measured by a vane-type anemometer in the following areas;
   1. At the floor level where the burner will be positioned during the test.
   2. 1.5 m (4.9 ft) above the enclosure floor where the cable tray will be positioned during the test.

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c. Wire ties Each specimen shall be separately attached to each rung of the cable tray.
d. Propane flow rate. 220 cm^3/s ± 8 cm^3/s when corrected to standard pressure and temperature (20 °C, 101 kPa)
   1. Theoretical heat output 20 kW (approx. 70,000 Btu/hr)
   2. Temperature 25°C ± 5°C
e. Air flow rate to the burner. Shall be 1280 cm^3/s ± 80 cm^3/s when corrected to standard pressure and temperature (20 °C, 101 kPa)
   1. Temperature of the air shall be 25 ± 5° C.
   2. Dew point of the air shall not be greater than 0 °C.

* 4.7.8.4 Observation. After burning has ceased, the cables shall be wiped clean and the longest charred or affected portion of the specimens measured and recorded to the nearest 25 mm (1 in). Cable damage shall be determined by measuring the distance of charring of the affected portion above the horizontal line from the lower edge of the burner face. The limit of charring shall be determined by pressing against the cable surface with a sharp object. Where the surface of the cable changes from a resilient surface to a brittle or crumbling surface the limit of charring has been determined. Distortion of the outer surface of the cable, such as blistering or melting immediately above the char shall be included in the damage measurement. On cable constructions which do not have charring, the limit for the affected portion shall be defined as "the point where the overall diameter is visibly reduced or increased".

Specimen failure shall be construed when any portion of any specimen burns to the top of the tray, either during or after operation of the ribbon burner. The following information (which does not constitute failure criteria) shall additionally be reported to help expand a performance data base: Flame temperature, period of time between burner shut off and cessation of flame on the specimen, overall distance of specimen jacket damage above the burner, overall distance of specimen conductor damage above the burner and a brief description of any dripping or flaming material which falls from the specimen to the floor during the test.

4.7.9 Gas flame. This test shall determine the ability of completed cable to maintain electrical circuit integrity in the presence of flame.

* 4.7.9.1 Specimen. The specimen shall consist of a 48 +/- 6 inch length of completed cable. The jacket and blocking materials shall be removed for a distance of 3 inches on each end of the sample. Sufficient insulation shall be removed from one end of each conductor for making electrical connections as follows:

(a) Specimens with one (1) conductor shall be tested with single phase voltage between conductor and armor with the armor grounded. For unarmored cables, a specimen shall be covered with either a copper braid or aluminum armor, providing no greater than 50 % coverage. The braid shall be attached to ground.
(b) Specimens with two (2) conductors shall be tested using two phases of the three phase source.
(c) Specimens with three (3) conductors shall be tested with each conductor connected to a separate phase.

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(d) Specimens with four conductors shall have physically opposite conductors and be connected together in pairs to form two effective electrical circuits.

(e) Specimens with six conductors shall have physically opposite conductors and be connected together in pairs to form three effective electrical circuits.

(f) Specimens with seven or more conductors shall have conductors connected together to form three effective electrical conductors. Connections shall be made such that physically adjacent conductors shall be members of different effective electrical conductors to the maximum practicable extent and such that the number of conductors constituting each effective electrical conductor does not differ by more than one.

* 4.7.9.2 Special apparatus. Apparatus shall include the following:

(a) A metered and adjustable source of natural gas or other flammable gas and an adjustable source of air (see figures 8 and 9), able to support the requirements as specified in 4.7.9.3. The gas and air support shall be made available as a mixture through a single conduit.

(b) A ribbon burner, which accepts gas and air from the gas and air supply via a single, 1/2 inch nominal pipe set vertically into its underside, which thoroughly blends the gas and air within an internal mixing chamber, which allows the blended gas and air mixture to exit for burning via a multiplicity of holes in its top which shall be fabricated in accordance with figures 4 and 4(a). The mixing chamber, fabricated from number 16 United States Standard Gage (USSG) steel, shall be horizontally centered within the burner and shall be airtight except for the specified holes on its two sides and through the inside of the 1/2 inch nominal pipe. The burner exterior shall be fabricated from number 14 USSG steel and shall be airtight except for the specified holes on its top and the specified holes in the mixing chamber.

(c) A 3 phase, 60 Hz voltage source, as shown in figure 11, shall provide a separate, single phase test voltage through a normal blow fuse or overload relay to each specimen conductor. This voltage source shall be in accordance with the following:

Specimens rated for 1 kV which incorporate conductors with cross sections of less than 14,000 circular mils and all specimens rated for less than 1 kV. The test voltage shall be 120 +/- 5 Vrms between conductors. Fuses or overload relays shall each have a maximum rating of 1 A.

The secondary winding voltage rating of any constituent voltage step up transformer shall be not less then the required test voltage. Each phase of the voltage source shall have the capacity to supply the specimen with a current which shall be not less than twice the fuse and overload relay rating.

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(d) A chamber enclosing the horizontally mounted burner and the suspended specimen (see 4.7.9.3). This enclosure shall incorporate an exhaust fan at its top and shall incorporate apertures for admitting air along its sides adjacent to its base. The exhaust fan shall be operated to produce no more suction than is necessary to carry off smoke and gases. The enclosure shall be designed to both minimize horizontal air drafts to the greatest practicable extent and to allow free circulation of convection air currents generated during the gas flame test.

4.7.9.3 Procedure. The burner shall be mounted within the chamber, such that the holes in its top are in common horizontal plane. The specimen shall be suspended within the enclosure, such that it is parallel to the burner, horizontally centered directly above the burner and spaced 1 7/8 +/- 1/8 inches vertically above the top of the burner surface. Specimen suspension shall be accomplished in a manner (such as by wire loops) which shall not significantly shield the specimen from flame and which shall prevent any portion of the specimen from dropping closer than 3/4 inch to the top surface of the burner during specimen burning. The gas air supply shall then be attached to the burner and the gas adjusted to provide a flow rate of not less than that calculated by the following formula:

Gas flow rate, in cubic feet per hour = 1387 (T+273)/PN

Where: T = Measured ambient temperature in the vicinity of the burner, in ^{o}C. Measurement accuracy shall be within +/- 0.5 ^{o}C.

P = Local atmospheric pressure, in $lb./in^2$ absolute. Measurement accuracy shall be within +/- 2 %.

N = Gross heating value, in $BTU's/ft^3$, of the gas at an atmospheric pressure of 14.69 $lb./in^2$ absolute and a temperature of 23°C, this value shall be accurate in +/- 2 %.

The gas exiting the burner shall then be ignited and the air flow to the burner adjusted to produce the smallest blue flame possible. The short circuit monitor test voltages shall then be applied to the specimen, as shown in figure 5. If the specimen is armored, an additional test voltage shall be applied by grounding the armor as shown in figure 5. (this will effectively apply an additional voltage which is approximately equal to the test voltage divided by the square root of 3 between the armor and the specimen conductors.) The flame and test voltages shall remain continuously applied to the specimen for a period of not less than that specified (see 3.1), unless specimen failure (see 4.7.9.4) occurs prior to the end of this period, in which case the test may be terminated.

4.7.9.4 Observation. Specimen failure shall be construed in any fuse blows or if any overload relay trips prior to completion of the specified test period (see 3.1).

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4.7.10 Heat distortion. This test shall determine the ability of insulation and jacket material to resist physical deformation when subjected to an elevated temperature.

* 4.7.10.1 Specimen. Each specimen shall be removed from completed cable in accordance with (a) and (b), whichever is indicated. The number of specimens which shall be required shall be in accordance with (b) (1), (2) or (3), as appropriate.

(a) Cable jacket: Each specimen shall consist of a piece jacketing that has been removed from a finished cable and a cut to form rectangular strip. This strip shall be not less than 7/8 inch long nor less than 9/16 ± 1/16 inch wide, and shall be ground and buffed to a thickness of 0.50 ± 0.010 inch with a grinding apparatus (see 4.7.10.2 (c)). The thickness of the specimen shall be made as uniform as practicable.

(b) Insulation: Each specimen shall consist of a single insulated conductor of length not less than 7/8 inch.

(1) Cables containing four or fewer conductors, one specimen shall be prepared from each insulated conductor.
(2) Cables containing more than four but fewer than 21 conductors, one specimen shall be prepared from each of four different insulated conductors (a total of four specimens).
(3) Cables containing 21 or more conductors, the number of specimens prepared shall be equal to the square root (rounded if necessary, to the nearest whole number) of the total number of conductors. Each specimen shall be prepared from a different insulated conductor.

* Note: In order to prevent misleading results, the conductor should be removed and replaced with a solid copper rod of similar diameter.

4.7.10.2 Special apparatus. Apparatus shall include the following:

(a) A self standing thickness gage, for supporting the requirements as specified in 4.7.10.3, which shall exhibit a measurement error of not less than $^{+}/_{-}$ 0.001 inch. This gage shall incorporate a flat horizontal platen upon which the entire specimen shall be laid flat during measurement and shall incorporate a foot which rests upon the top of the specimen during measurement. This foot shall be attached to a plunger which moves freely in the vertical direction and which is attached to a dial or other indicating device from which the specimen thickness may be read. The foot shall present a horizontal disk of 3/8 + 0 minus 1/64 inch diameter to the specimen, which shall bear down on the specimen with a force of 85 + 0, - 4 grams and shall be loaded with weights (see 4.7.10.3) to present additional force to the specimen.

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(b) A micrometer (required for insulated conductor specimens only) with flat, parallel measurement surfaces on both spindle and anvil, for the specimen thickness measurement as specified in 4.7.10.3 (b)(1), which shall exhibit a measurement error of not more than +/- 0.001 inch.

(c) A motor driven grinding wheel and a motor driven buffing wheel, or their equal, for the specimen preparation as specified in 4.7.10.1 (a) and (b)(1). Guides shall be provided to assure that the specimen is pulled tangentially to the surface of each wheel during grinding and buffing.

(d) An oven, for accommodating the gage with the specimen inserted and for supporting the requirements as specified in 4.7.10.3. The oven air temperature shall be measured in the immediate vicinity of the specimen.

4.7.10.3 Procedure. The initial insulation thickness of each specimen (T in the formula below) shall be determined (see (a)(1) and (b) below) and the oven shall be preheated to 121 +/- 1°C. This oven temperature shall be maintained for the remainder of the test. The gage foot shall then be loaded with weights (see (a)(2) and (b)(2) below) and the gage placed within the oven. Not less than one hour later, one specimen shall be selected and also placed within the oven. Not less than one hour after placing the specimen within the oven, the specimen shall be placed beneath the gage foot such that the gage indicates the specimen thickness. Not less than 1 hour after placing the specimen beneath the gage foot, the final specimen thickness or diameter (as appropriate) shall be read from the gage and the final insulation thickness (t in formula below) determined (see (a)(3) and (b)(3) below). In an identical manner, the remaining specimens shall also be tested. The percentage heat distortion of the insulation shall then be calculated by using the following formula:

Percentage heat distortion = Median value of 100(T-t)/T for all tested specimens.

Where: T = Initial insulation thickness of the specimen, in thousandths of an inch (see (a) and (b) below).

t = Final insulation thickness of the specimen, in thousandths of an inch (see (a) and (b) below).

The following additional details apply:

(a) Rectangular strip specimens:

(1) The initial insulation thickness (T) shall be the initial specimen thickness, as measured with the gage, prior to loading the gage with weights and prior to inserting either the gage or the specimen into the oven

(2) The gage foot shall be loaded with 2000 +/- 100 grams.

(3) The final insulation thickness (t) shall be the final specimen thickness, measured as specified.

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(b) Insulated conductor specimen:

(1) The initial insulation thickness (T) shall be defined as 1/2 of the difference between the measured overall specimen diameter and the measured specimen conductor diameter. Measurements shall be made using the micrometer. The overall specimen diameter shall be measured in the same radial direction as will be measured when the final insulation thickness (t) (see (b)(3) below) is determined.

(2) The gage foot shall be loaded as follows:
   a. Specimens with a conductor cross section of not more than 3999 circular mils nor less than 2300 circular mils: 400 +/- 20 grams.
   b. Specimens with a conductor cross section of not more than 21,999 circular mils nor less than 4000 circular mils: 500 +/- 25 grams.
   c. Specimens with a conductor cross section of not more than 94,999 circular mils nor less than 22,000 circular mils: 750 +/- 35 grams.

(3) The final insulation thickness (t) shall be defined as 1/2 of the difference between the final specimen diameter, measured as specified and the previously measured specimen conductor diameter.

4.7.10.4 Observation. The specimen shall be construed to have failed the test if the percentage heat distortion is greater than that specified on the applicable specification sheet.

4.7.11 Permanence of printing (conductor insulation). This test shall determine the ability of printed information on insulated conductors to remain legible in the presence of repeated abrasion.

4.7.11.1 Specimen. Each specimen shall consist of a single insulated conductor which shall have sufficient length for use in the test as specified in 4.7.11.3. Specimens shall be removed from completed cable as follows:

(a) Cables containing four or fewer conductors: one specimen shall be removed from each insulated conductor.

(b) Cables or cords containing more than four but fewer than 21 conductors: one specimen shall be removed from each of four different insulated conductors (a total of four specimens).

(c) Cables or cords containing 21 or more conductors: The number of specimens removed shall be equal to the square root (rounded, if necessary, to the nearest whole number) or the total number of conductors. Each specimen shall be removed from a different insulated conductor.

4.7.11.2 Special apparatus. Apparatus shall include an abrading machine, which shall secure a specimen horizontally between two fixed supports, and which shall abrade the specimen printed identification by means of a motor-driven, reciprocating tape consisting of number 50-2/20 unbleached cotton braid. One possible arrangement for the abrading machine is shown on figure 12. The braided cotton tape shall lie flat on the specimen surface, shall remain perpendicular to the specimen axis, and shall abrade an unchanging area of the specimen periphery, over an

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arc of not less than 135 degrees, at all times during the test. One end of the tape shall be reciprocated at a rate of 28 ± 3 cycles per minute, such that the tape is drawn across the specimen for a distance of 10 plus 1, minus 0 inches in each direction (20 inches minimum total excursion during cycle). The opposite end of the tape shall be maintained under tension by a freely suspended 1/2 plus 1/64, minus 0 pound weight. Each side of the tape shall be used to perform not more than 500 cycles; the tape shall then be discarded and replaced. The abrading machine shall incorporate an automatic counter to total the number of abrasion cycles to which the specimen is subjected during the test.

4.7.11.3 Procedure. Each specimen shall be wiped with a clean, dry cloth to remove any lubricant or dirt. One specimen shall then be selected and mounted between the two fixed supports of the abrading machine. Mounting shall be accomplished such that the specimen shall be as taut as practicable, and such that the braided cotton tape shall repeatedly abrade the same portion of the specimen printed identification at all times during the test. The automatic counter shall be set initially to zero. The abrading machine motor shall then be turned on. Allowing the braided cotton tape to reciprocate and abrade the specimen printed identification either until the printed identification is no longer legible in its abraded region, or until not fewer than the specified number of abrasive cycles (see 3.1) have been completed, whichever occurs first. The number of abrasive cycles completed shall then be noted, and the remaining specimens then tested in an identical manner.

4.7.11.4 Observation. Specimen failure shall be construed if the median value of the number of abrasive cycles completed on each specimen is less than that specified (see 3.1).

4.7.12 Permanence of printing (jacket). This rest shall determine the ability of printed information of jacketing material to remain legible in the presence of repeated abrasion.

4.7.12.1 Specimen. The specimen shall consist of a piece of completed cable which shall have sufficient length for use in the test as specified in 4.7.12.3.

4.7.12.2 Special apparatus. Apparatus shall include an abrading machine which shall support and secure the specimen such that it is straight and horizontal, and which shall abrade the specimen printed identification by means of a motor driven, transversely reciprocating steel pin. This steel pin shall have a diameter of 0.025 ± 0.001 inch where it abrades the specimen, and shall have a surface roughness of not less than 2 micro inches, as specified in ANSI B46.1. (A selected sewing needle may satisfy these requirements.) One possible arrangement for the abrading machine is shown on figure 13. The steel pin shall be horizontal and perpendicular to the specimen axis, shall ride along the top of the specimen had shall be weighted to bear down on the specimen with a force of 1 plus 1/16, minus 0 pound at all times during the test. This pin shall be reciprocated at a rate of 60 ± 2 cycles per minute, such that the pin is drawn along the specimen for a distance of 3/8 plus 1/16, minus 0 inch in each direction (3/4 inch minimum total excursion) during each cycle. The abrading machine shall incorporate an automatic

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counter to total the number of times that the specimen is abraded by the steel pin during the test.

4.7.12.3 Procedure. The specimen shall be wiped with a clean, dry cloth to remove any lubricant or dirt, and shall be secured in the abrading machine with the specimen printed identification facing upwards, where it is to be abraded by the steel pin. The automatic counter shall be set initially to zero. The abrading machine motor shall then be turned on, allowing the steel pin to reciprocate and abrade the specimen printed identification either until the printed identification is no longer legible in the abraded region, or until not fewer than 250 abrasive cycles have been completed, whichever occurs first. The number of abrasive cycles completed shall then be noted. This test shall be repeated four more times (five times total), subjecting a fresh portion of the specimen printed identification to abrasion each time.

4.7.12.4 Observation. Specimen failure shall be construed if the median value of the number of abrasive cycles completed during each of the five tests is less than that specified (see 3.1).

4.7.13 Physical tests on jacket. This test shall determine whether or not various jacketing materials have been properly processed, by means of tension measurements. This test makes provision for making tension measurements both before and after these materials have been artificially aged.

4.7.13.1 Specimens. Each specimen shall consist of a single piece of jacketing, which shall have sufficient length for use in the test as specified in 4.7.22.

4.7.13.2 Special apparatus. Apparatus shall include the following:

(a) A forced fresh air circulating oven for supporting the requirements as specified in 4.7.13.3(a). The oven air temperature shall be measured in the immediate vicinity of specimens.
(b) A heated container, which shall be filled with lubricating oil in accordance with symbol number 2190 TEP of MIL-L-17331, and which shall be for the specimen immersion as specified in 4.7.13.3(b). The oil shall be maintained at a temperature of 121 ± 1°C during specimen immersion.

4.7.13.3 Procedure. Specimens shall be maintained at ambient temperature for a period of not less than 30 minutes immediately prior to any accelerated aging or testing. Each of the specified physicals (aged) accelerated aging procedures shall be performed as follows, using one specimen for each procedure. The same specimen shall not be used for more than one physicals (aged) procedure. Each of the specified physicals (unaged) test shall be performed as specified in 4.7.22.

(a) Air oven accelerated aging: each specimen shall be freely suspended vertically, secured by one end, within the oven. The air temperature within the oven shall then be raised to 136 ± 2° C and shall be maintained at this value for a continuous period of not less than 168 hours. Each specimen shall then be removed from the oven and tested as specified in 4.7.22.

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(b) Hot oil immersion accelerated aging: each specimen shall be submerged within the hot oil bath for a continuous period of not less than 18 hours. Each specimen shall then be removed from the hot oil bath, blotted lightly to remove excessive oil and then suspended in air, to ambient room temperature, for 4 ±1/2 hours. Each specimen shall then be tested as specified in 4.7.22.

4.7.13.4 Observation. Any of the following shall constitute specimen failure:

(a) A specimen tensile strength which is less than that specified (see 3.1).
(b) A specimen elongation which is less than that specified (see 3.1).

4.7.14 Tear strength. This test shall determine the ability of elastomeric jacketing material to withstand the propagation of a cut passing through a portion of its length.

4.7.14.1 Specimen. Each specimen shall consist of a piece of jacketing material, which shall have been cut to the dimensions specified in ASTM D 470. Specimens shall be removed from completed cabled. Not fewer than five specimens shall be required.

4.7.14.2 Special apparatus. Apparatus shall include the following:

(a) A micrometer with flat, parallel measurement surfaces on both spindle and anvil, which shall be for the specimen measurements as specified in 4.7.22.3 and which shall exhibit a measurement error of not more than plus or minus 0.001 inch.
(b) A motor driven tensile machine which shall be for applying increasing tension to the specimen, and which shall automatically indicate within plus or minus 1 percent the maximum tension experienced by the specimen prior to tearing. This tensile machine shall incorporate two parallel and opposing jaws, between which the specimen can be secured, as specified in 4.7.14.3. The two jaws shall increase their separation at the uniform rate of 20 +/- 2 inches per minute by means of the tensile machine motor.

4.7.14.3 Procedure. The specimen shall be tested in accordance with ASTM D 470.

* 4.7.14.4 Observation. Specimen failure shall be construed if the median value of the measured tear strengths of all specimens is less than that specified (see 3.1).

4.7.15 Watertightness. This test shall determine the ability of completed cable, which is intended to pass through watertight bulkheads, to prevent the longitudinal flow of low-pressure water.

4.7.15.1 Specimen. The specimen shall consist of a 60 ± 2 inch length of completed cable. The specimen shall be cut to length using a scissors-action cable cutter. (Saws shall not be used.)

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4.7.15.2 Special apparatus. Apparatus shall include the following:

(a) A source of pressurized water, which shall be provided at a regulated pressure of 25 ± 1/2 lb/in^2 and which shall be for use with the terminal fitting (see (b)) as specified in 4.7.15.3.

(b) A metal terminal fitting which applies the source of pressurized water to one end of the specimen, which supports the requirements as specified in 4.7.15.3, and which shall be fabricated as specified in the following. Figure 14 shows one possible arrangement for the terminal fitting. The fitting shall admit the specimen end for the distance specified in 4.7.15.3, and shall have an i.d., where it fits over the specimen, of not greater than the measured overall specimen diameter plus 1/2 inch. The fitting shall have a means for introducing the source of pressurized water to the specimen end, and means for bleeding off any air which might be trapped between the specimen end and the source of pressurized water. The fitting shall also have an aperture for introducing a hardening sealant (see (c)), to produce a pressure-tight bond between the fitting and the specimen jacket. A plug (such as a thick wrapping of rubber tape around the specimen) shall be provided at the place where the specimen enters the fitting, to prevent sealant loss and to approximately center the specimen within the fitting.

(c) A liquid hardening sealant which shall be for producing a pressure tight bond between the fitting and the specimen jacket, when applied as specified (see 4.7.15.3) and allowed to harden. The sealant shall be at a temperature of not greater than 100°C when poured into the terminal fitting, and shall not expand in volume while hardening. An appropriate metal alloy is recommended as a sealant since it can be remelted and reused. Any metal alloy used shall have a melting point of not greater than 88°C.

4.7.15.3 Procedure. The specimen shall be secured in a vertical position, the terminal fitting (including the plug) shall be fitted over its upper end, and the hardening sealant shall be poured through the appropriate terminal fitting aperture to bond the fitting to the specimen. These shall be done such that the specimen is approximately centered within the terminal fitting, such that not more than a 6 inch length of specimen jacket is exposed to the sealant, such that the end face of the specimen is located a distance of not more than 1 inch above the sealant surface and itself contains no sealant, and such that the aperture through which the sealant was introduced is itself made pressure tight. The sealant shall be allowed to harden, following which the terminal fitting plug shall be removed or cut away to the extent that it exerts no radial compression on the specimen. The specimen shall then be positioned such that it is straight, and such that no portion of the specimen is at a greater elevation than the specimen end within the terminal fitting. The fitting shall then be connected to the source of pressurized water, and excess air within the fitting shall be bled off until it can be certain that the entire enclosed specimen end face is exposed to the pressurized water for the remainder of the test.
The source of pressurized water shall remain connected to the fitting for a continuous period of 6 plus 1/4, minus 0 hours. During this test period,

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water (if any) leaking from the specimen end opposite the terminal fitting shall be collected into an initially dry container. Following the test period, the volume of this collected water shall be measured.

4.7.15.4 Observation. Specimen failure shall be construed if the collected water volume exceeds that shown in table XI.

TABLE XI. Limits for water leakage.

Sum of specified cross sections of all specimen conductors (see 3.1) (circular mills)	Collected water volume (cubic inches, maximum)
Less than 9,001	4
9,001 to 15,000	5
15,001 to 25,000	6
25,001 to 50,000	7
50,001 to 100,000	8
100,001 to 200,000	10
200,001 to 500,000	11
500,001 to 800,000	12
800,001 or greater	13

4.7.16 Acid gas generation. The method given below is to determine the total emission of any strong soluble acids (pH less than 3). The required apparatus is shown on figure 14. A weighed sample of the jacket, fillers and insulation materials (for component wire, use insulation removed from a 22 AWG wire), normally 1/4 to 1/2 gram, shall be placed in a silica boat which is put into the center of a silica tube, length 50/60 centimeters and internal diameter 20/22 millimeters. The materials shall be tested individually. The silica tube shall be placed in the tube furnace. An air supply, derived from a blower or compressed air cylinder, at the rate of 1 liter per minute plus or minus 5 percent shall be passed through the silica tube and then through four absorber flasks each containing 30 milliliters of deionized water. The furnace heating shall be commenced and the temperature of the tube and sample shall be raised to 800 ± 10°C over a period of approximately 40 minutes and then held at temperature for a further 20 minutes. During the heating period, any acid gases produced will be carried over into the absorber flasks. on completion of the heating cycle, the acid content of the fluids in the absorber flasks shall be titrated against 0.1 N sodium hydroxide solution using congo red as an indicator. The total titre indicates the total soluble acids. 1.0 ml of 0.1 N sodium hydroxide solution is equivalent to 3.65 mg of acid expressed as "acid equivalent relative to hydrochloric acid."

4.7.16.1 Observation. Specimen failure shall be construed if the acid equivalent exceeds the percentage, by weight of the sample, specified (see 3.1).

4.7.17 Halogen content. The halogen content of the cable jacket or fillers shall be determined by X-ray fluorescence or analytically following an examination and analyses of the chemical composition of all ingredients used.

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4.7.17.1 Observation. Specimen failure shall be construed if the halogen content is greater than that specified (see 3.1).

* 4.7.18 Smoke index. The smoke index shall be measured as specified in NES 711 with exceptions and modifications as listed in 4.7.18.1 AND 4.7.18.2

4.7.18.1 Specimen.

(a) Jacket. Each specimen shall consist of a sufficient number of 75 mm long strips cut from the cable jacket to completely cover the face area of the sample holder. To prevent excessive buckling and distortion of the specimen during test, a wire mesh, manufactured from either A 1.0 mm diameter stainless steel wire with a spacing of 12.5 mm and a square mesh configuration or a 40 MIL stainless steel wire 2 mesh screen (2 openings per inch), shall be placed inside and across the face of the sample holder. The specimen shall be fabricated by placing the specimen holder (with wire mesh) test face down onto a flat surface and positioning each 75 mm length in the holder in a parallel arrangement so that when the holder is in the test position the strips will be vertical. An insulating block 10 mm thick completely wrapped in aluminum foil shall be placed on top of the strips followed by a tension spring and locking pin.

(b) Insulation. The test specimen shall be a 1 meter length of 22 AWG finished wire with a nominal wall thickness of 0.020 inch of insulation material.

* (c) Fillers and sealants.

1. NON FIBROUS The test specimen shall be 3 by 3 inches of 0.070 ± 0.010 inch thick block of the same material used in the cable.
2. FIBROUS The test specimen shall consist of as many 75 mm lengths of 0.070 ± 0.010 inch diameter strands necessary to fill the 3 by 3 inch test specimen holder one layer deep and as closely spaced as possible. The test specimens shall be prepared as described in 4.7.27.1 (A)

* 4.7.18.2 Special procedures. Special procedures shall include the following:

A. Only the use of propane gas shall be allowed.
B. The Radiometer shall be used to calibrated the heat flux at the initial start of each test period.
C. The test chamber shall be calibrated using a NIST CELLULOSE standard at the initial start of each test period.
D. The chamber shall be inspected to assure airtightness. Seals and gaskets shall be replaced as needed.
E. The test shall be run using a minimum of 3 specimens. If test values are below the limits specified (see 3.1) test is complete. If one test value deviates from the median value by more than 20 %, two additional tests shall be performed. The high and low test values shall be discarded and the remaining 3 averaged.

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4.7.18.3 Observation. Specimen failure shall be construed if any smoke index is greater than that specified (see 3.1).

4.7.19 Immersion tests. Specimens of the cable jacket material (see 4.7.13.1) shall be immersed in the fluids shown in table XII for 24 hours at the temperatures specified. The specimens shall then be removed, blotted to remove excess fluid, then suspended in air at room temperature for not less than 3-1/2 nor more than 4-1/2 hours. Each specimen shall be tested as specified in 4.7.22.

TABLE XII. Immersion test temperatures.

Fluid	Test temperature
Fuel oil, MIL-F-16884	98 to 100°C
Turbine fuel, JP-8, MIL-T-5624	48 to 50°C
Cleaner, Isopropyl alcohol, TT-I-735	20 to 25°C
Hydraulic fluid, MIL-H-5606	48 to 50°C
Hydraulic fluid, MIL-H-17672	48 to 50°C
Lubricating oil, MIL-L-23699	98 to 100°C
Coolant, Monsanto Coolanol 25 or equivalent	20 to 25°C

4.7.19.1 Observation. Any of the following shall construe specimen failure:

(a) A specimen tensile strength which is less than that specified (see 3.1).
(b) A specimen elongation which is less than that specified (see 3.1).

4.7.20 Toxicity index. The toxicity index shall be determined as specified in NES 713. For insulation materials, the toxicity index shall be calculated on the mass of insulation on 1 meter length of finished wire size 22 AWG with a nominal wall thickness of 0.020 inch of insulation material. For jacket and sealant materials, the toxicity index shall be calculated on 100 grams of material. (The toxicity index is derived from the chemical analysis of the products of combustion of the materials.)

4.7.20.1 Observation. Specimen failure shall be construed if any toxicity index is greater than that specified (see 3.1).

4.7.21 Weathering. Jacket material shall be tested in accordance with ASTM D 2565. The following conditions shall apply:

Xenon arc lamp, 6500 Watt, Borosilicate glass filters
Irradiance: 1.75 w/m^2 .nm at 500 nm
Exposure: Arc lamp on: 18 hours, Black panel temp 50 ± 2°C
Relative humidity 50 ± 2 percent
(for 18 minutes every 2 hours water is sprayed onto specimen)
Arc lamp off: 6 hours
Temperature 25 ± 2°C
Relative humidity 90-95 percent
Total exposure time: 1000 hours.

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4.7.21.1 Observation. Specimen failure shall be construed if the tensile strength and elongation retention is less than 75 percent or if the specimen surface exhibits signs of cracking when examined using 3X magnification.

4.7.22 Tensile strength and elongation. Both aged and unaged specimens of the insulation and jacket material shall be tested in accordance with methods 3021 and 3031 of FED-STD-228. Unless otherwise specified on the applicable specification sheet, there shall be 1 inch bench marks, 1 inch jaw separation and a rate of jaw travel of 10 inches per minute. For cross linked polyethylene insulation meeting the criteria of ASTM D 1248 for type II, III and IV insulation the rate of jaw travel shall be 2 inches per minute. The thickness of the specimen shall be measured using any suitable micrometer.

4.7.22.1 Observation. Specimen failure shall be construed if the tensile strength and elongation retention is less than that specified herein or in the applicable specification sheet.

4.7.23 Shrinkage. A 12 inch specimen of cable shall be cut so that all components are flush at both ends. Unless otherwise specified in the applicable specification sheet, the specimen shall then be aged at 136 ± 3°C for 6 hours in an air circulating oven. At the end of this period, the specimen shall be removed from the oven and allowed to cool to room temperature. Shrinkage of the jacket shall then be measured to determine the total distance the jacket has receded from both ends of the conductor.

4.7.23.1 Observation. Specimen failure shall be construed if the total shrinkage of the jacket is greater than 0.25 inch.

4.7.24 Weight. The weight of each lot of finished cable shall be determined as specified in procedure I (see 4.7.24.1). Lots failing to conform to the weight requirement specified in the applicable specification sheet, when tested in accordance with procedure I shall be subjected to procedure II (see 4.7.24.2). Reels or spools failing to conform to the requirements specified in the applicable specification sheet when tested by procedure II shall be rejected.

4.7.24.1 Procedure I. The length and weight of a specimen not less than 10 feet long shall be accurately measured and the resultant measurements converted to pounds per 1000 feet.

4.7.24.2 Procedure II. The net weight of the finished cable on each reel or spool shall be determined by subtracting the tare weight of the reel or spool from the gross weight of the reel or spool containing the finished cable. The net weight of cable on each reel or spool shall be divided by the determined length of finished cable on that reel or spool and the resultant measurement converted to pounds per 1000 feet. When wood or other absorbent materials are used for reel or spool construction, weight determinations shall be made under uniform conditions of relative humidity.

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* 4.7.25 Cross linking verification. Cable jacket cross link verification shall utilize the hot creep test in accordance with ICEA T-28-562. The procedure for jacket testing shall be modified to run at 200 °C. Jacket percent of elongation shall be as specified in the individual slash sheets (see 3.1)

4.8 Test methods (electrical).

4.8.1 Attenuation. This test shall determine the ability of completed cable, which is intended for the transmission of data, voice or control information, to propagate a signal at a frequency of 3 megahertz (MHz) without causing an unacceptable reduction of signal amplitude.

4.8.1.1 Specimen. The specimen shall consist of a piece of completed cable, which shall have sufficient length to exhibit an electrical attenuation of not less than 3 decibels (dB) at the specified test frequency (see 3.1). Individual transmission lines within this specimen (where each single insulated conductor plus surrounding shield or each insulated conductor pair plus surrounding shield is considered to be a transmission line) shall be selected for testing as follows.

(a) Specimens containing four or fewer transmission lines: every transmission line shall be tested.
(b) Specimens containing more than four but fewer than 21 transmission lines: Four different transmission lines shall be tested. Not fewer than one of these transmission lines shall be selected from each concentric cabling layer within the specimen.
(c) Specimens containing 21 or more transmission lines: The number of transmission lines tested shall be equal to the square root (rounded, if necessary, to the nearest whole number) of the total number of transmission lines. Not fewer than one of these transmission lines shall be selected from each concentric cabling layer within the specimen.

An electrical connector, which shall be designed for use at the specified test frequency (see 3.1), and which shall be appropriate for use with the specimen, shall be attached to each end of each transmission line selected for testing.

4.8.1.2 Special apparatus. Apparatus shall include the following, and shall be used at the specified test frequency (see 3.1), and shall be chosen such that all electrical interconnections (see 4.8.1.3) shall be of the same nominal characteristic impedance throughout (except that connectors and specimens in which a single conductor plus surrounding shield is tested may be of a different impedance than this apparatus, if necessary).

(a) A sinusoidal signal generator, which shall produce an unchanging signal of the specified test frequency (see 3.1) within a frequency accuracy of 0.25 percent. The signal generator may be amplitude modulated in a constant and unchanging manner if the detector (see (b)), requires an amplitude modulated signal for proper operation. The signal generator shall generate power to produce a reasonable indication on the detector when used as

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specified in 4.8.1.3, however, the signal generator shall not generate so much power that the specimen or any piece of test apparatus will consequently experience a significant temperature rise.

(b) A crystal diode, bolometer or other detector, with its associated indicating device, which shall be used as specified in 4.8.1.3.

(c) A calibrated variable attenuator, which shall produce a maximum attenuation greater than that exhibited by the specimen, and which shall resolve attenuation within an accuracy of plus or minus 0.25 dB.

(d) Two fixed attenuators, each of which shall exhibit an attenuation of not less than 10 dB.

(e) Impedance matching transformers (required only when testing a conductor pair plus surrounding shield): Two, two winding transformers; one winding of each transformer shall have an impedance equal to the nominal impedance of the fixed attenuators; the second winding of each transformer shall be center tapped, and shall have an impedance equal to the specified impedance of the specimen (see 3.1). When testing a single conductor plus surrounding shield, impedance matching transformers may be used, if necessary, to match the nominal unbalanced impedance of each fixed attenuator (see (d)), to the specified unbalanced impedance of the specimen (see 3.1), as shown on figure 16.

(f) An adapter connector (may not be required, see 4.8.1.3) for joining the two fixed attenuators directly together. This adapter connector shall exhibit an attenuation which shall be not greater than 2 percent of the specimen attenuation. If desired, the adapter connector may be fabricated in the same manner and of the same components as the specimen, except that the shortest practicable length of completed cable or cord shall be used to join the two connectors.

4.8.1.3 Procedure. The test apparatus shall be electrically interconnected as shown on figure 11 (when the transmission line to be tested is a single insulated conductor plus surrounding shield) or as shown on figure 17 (when the transmission line to be tested is an insulated conductor pair plus surrounding shield), as applicable; one of the selected transmission lines shall then be connected where shown. Following this, the calibrated variable attenuator shall be adjusted to a low value, and the signal generator adjusted to produce a test signal of sufficient amplitude to produce a reasonable indication on the detector. The setting on the calibrated variable attenuator and the indication on the detector shall then be noted. Next, the transmission line shall be disconnected from the test apparatus, and the two fixed attenuators shall be connected together either directly, or, if necessary, by means of an adapter connector. The signal generator shall not be readjusted. The calibrated variable attenuator shall then be readjusted until the detector produces the same indication as was noted when the transmission line was connected to the test apparatus. The new setting on the calibrated variable attenuator shall then be noted, and the transmission line attenuation shall be calculated by using the following formula:

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Attenuation, in dB per 100 feet = 100 $(A_2 - A_1)/L$

Where: A_1 = Initial setting of the calibrated variable attenuator (when the transmission line is connected to the test apparatus), in dB.
A_2 = Final setting of the calibrated variable attenuator (following transmission line removal from the test apparatus), in dB.
L = Measured specimen length, in feet; measurement accuracy shall be within plus or minus 1 percent.

In an identical manner, the remaining selected transmission lines shall also be tested.

4.8.1.4 Observation. Specimen failure shall be construed if any transmission line exhibits an attenuation which is greater than that specified (see 3.1).

4.8.2 Capacitance. This test shall determine the capacitive characteristics of completed cable.

4.8.2.1 Specimen. The specimen shall consist of a 122 ± 1/2 inch length of completed cable, which shall have overall components (such as armors, jackets or overall binders) removed from each end for a distance of 2 plus 1/2, minus 0 inches, thereby exposing the insulated conductors (and shields, if applicable) at each end of the cable core. Each exposed shield (if any) at each cable end shall be unwrapped from the end of its enclosed insulated conductor(s) for a distance of 1 ± 1/8 inch; the unwrapped portion of each shield shall then be tightly twisted to form a pigtail for electrical connection. The insulation shall next be stripped from each end of each insulated conductor for a distance of 1/2 ± 1/8 inch. If the characteristic impedance test is not required (see 3.1), then the insulation may be stripped, for the specified distance from all of the insulated conductors at one specimen end only.

4.8.2.2 Special apparatus. Apparatus shall include an instrument for measuring capacitance (such as a capacitance bridge, Q-meter or other instrument) which shall be for supporting the requirements as specified in 4.8.2.3. This apparatus shall have a two terminal measurement port, which shall use a sinusoidal wave of the specified frequency (see 3.1) for making measurements, and shall make measurements within an accuracy of plus or minus 1 percent.

4.8.2.3 Procedure. The appropriate capacitance test method shall be selected and performed from the following:

(a) Specimens incorporating any number of individually shielded, insulated conductor transmission lines: One transmission line shall be selected for measurement, and each end shall be positioned such that its conductor and shield shall not come into mutual electrical contact, nor into electrical contact with any other conductor or shield which may be within the specimen. One end of this selected transmission line shall then be appropriately connected electrically to the capacitance measuring

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instrument, and its capacitance shall be measured. In an identical manner, the capacitance of every other transmission line within the specimen shall also be measured. The capacitance per foot of each transmission line shall then be calculated by using the following formula:

Capacitance in picofarads per foot = C/10

Where: C = Measured capacitance of the transmission line, in picofarads.

(b) Specimens incorporating any number of individually shielded, insulated conductor pair transmission lines: one transmission line shall be selected for measurement, and each end shall be positioned such that neither its conductors nor its shield come into mutual electrical contact, or into electrical contact with any other conductor or shield which may be within the specimen. Three capacitance's shall then be measured at one end of this transmission line as follows, where one conductor of this line has been designated as conductor number 1 and the other conductor of this line has been designated as conductor number 2 as follows:

(1) Conductor number 1 and the shield shall both be electrically connected to one terminal of the capacitance measuring instrument, and conductor number 2 shall be electrically connected to the other instrument terminal. The capacitance shall then be measured and designated as C_a.
(2) Conductor number 2 and the shield shall next be electrically connected to one terminal of the capacitance measuring instrument, and conductor number 1 shall be electrically connected to the other instrument terminal. The capacitance shall then be measured and designated by C_b.
(3) Both conductor number 1 and conductor number 2 shall then be electrically connected to one terminal of the capacitance measuring instrument, and the shield shall be electrically connected to the other instrument terminal. The capacitance shall then be measured and designated as C_c. The mutual capacitance per foot and, if specified (see 3.1), the capacitance unbalance (sometimes called the coefficient of asymmetry) shall then be calculated by using the following approximate formulas:

Mutual capacitance, in picofarads per foot = $(C_a + C_b)/20 - C_c/40$

Capacitance unbalance, in percent = $10|C_a - C_b|/C_m$

Where: C_a, C_b and C_c are as defined in (1), (2) and (3) respectively, in picofarads.
C_m = Calculated mutual capacitance, in picofarads per foot.

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In an identical manner, the mutual capacitance and, if specified (see 3.1), the capacitance unbalance of every other transmission line shall also be measured and calculated.

(c) Specimens incorporating multiple, unshielded, insulated conductor pair transmission lines: the mutual capacitance and, if specified (see 3.1), the capacitance unbalance of every transmission line shall be determined, using the same method as used for specimens incorporating individually shielded, insulated conductor pair transmission lines, with the following change: the overall specimen braid (if any) and all specimen conductors except the conductor pair under test shall be connected together electrically and considered to be the shield of the transmission line under test.

(d) Specimens incorporating any number of individually shielded, insulated conductor triad transmission lines: one transmission line shall be selected for measurement, and each end shall be positioned such that none of its conductors, nor its shield, come into mutual electrical contact, nor into electrical contact with any other conductor or shield which may be within the specimen. Four capacitances shall then be measured at one end of this transmission line as follows, where one conductor of this line has been designated as conductor number 1, another conductor of this line has been designated as conductor number 2, and the remaining conductor of this line has been designated as conductor number 3 as follows:

(1) Conductors number 2 and 3 and the shield shall be electrically connected to one terminal of the capacitance measuring instrument, and conductor number 1 shall be electrically connected to the other instrument terminal. The capacitance shall then be measured and designated as C_a.

(2) Conductors number 1 and 3 and the shield shall next be electrically connected to one terminal of the capacitance measuring instrument, and conductor number 2 shall be electrically connected to the other instrument terminal. The capacitance shall then be measured and designated as C_b.

(3) Conductors number 1 and 2 and the shield shall next be electrically connected to one terminal of the capacitance measuring instrument, and conductor number 3 shall be electrically connected to the other instrument terminal. The capacitance shall then be measured and designated as C_c.

(4) Conductors number 1, 2 and 3 shall then be electrically connected to one terminal of the capacitance measuring instrument, and the shield shall be electrically connected to the other instrument terminal. The capacitance shall then be measured and designated as C_d.

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The mutual capacitance per foot and, if specified (see 3.1), the capacitance unbalance (sometimes called the coefficient of asymmetry) shall then be calculated by using the following approximate formulas:

a.
$$C_m = \frac{3(C_a + C_b + C_c) - C_d}{12 \times (\text{length of Specimen in feet})}$$

b. The capacitance unbalance or coefficient of asymmetry (K) of a shielded triad, expressed in percent, shall be determined by the formula:

$$K_1 = 10^{\circ}C_a - C_b^{\circ}/C_m$$

$$K_2 = 10^{\circ}C_a - C_c^{\circ}/C_m$$

$$K_3 = 10^{\circ}C_b - C_c^{\circ}/C_m$$

Where: K_1 = The percent of capacitance unbalance of conductor No. 1 in relation to conductor No. 2.
K_2 = The percent of capacitance unbalance of conductor No. 2 in relation to conductor No. 3.
K_3 = The percent of capacitance unbalance of conductor No. 3 in relation to conductor No. 1.

In an identical manner, the mutual capacitance and, if specified (see 3.1), the capacitance unbalance of every other transmission line shall also be measured and calculated.

(e) The mutual capacitance and, if specified (see 3.1), the capacitance unbalance of every transmission line shall be determined, using the same method as used in (d), for specimens incorporating individually shielded, insulated conductor triad transmission lines with the following change: the overall specimen braid (if any) and all specimen conductors except the conductor triad under test shall be connected together electrically and considered to be the shield of the transmission line under test.

4.8.2.4 Observation. Specimen failure shall be construed if the specimen exhibits a mutual capacitance or unbalance which is greater than that specified (see 3.1).

* 4.8.2.5 If the characteristic impedance and/or mutual inductance tests are specified (see 3.1), then the specimen shall be retained for use in that test.

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4.8.3 Characteristic impedance. This test shall determine the characteristic impedance encountered by signals propagating along completed cable. The mutual capacitance (see 4.8.2) shall be calculated for each transmission line within the completed cable or cord prior to performing the characteristic impedance test. This is necessary since values of mutual capacitance are used to calculate values of characteristic impedance.

4.8.3.1 Specimen. The specimen shall be that which was previously used to determine capacitance characteristics (see 4.8.2.1).

4.8.3.2 Special apparatus. Apparatus shall include an instrument for measuring inductance (such as an inductance bridge, Q meter or other instrument) which shall support the requirements as specified in 4.8.8.3. This apparatus shall have a two terminal measurement port, which shall use sinusoidal wave of the specified frequency (see 3.1) for making measurements. Measurements shall be made within an accuracy of plus or minus 1 percent.

4.8.3.3 Procedure. Select and perform the appropriate characteristic impedance test method from the following:

(a) Specimens incorporating any number of individual shielded, insulated conductor transmission lines shall be laid straight. At one specimen end the exposed conductor of each transmission line shall be electrically connected to the pigtail of its associated shield by tightly twisting both together. At the opposite specimen end, one transmission line shall be selected for measurement, which shall then be connected to the inductance measuring instrument. The inductance of this transmission line shall be measured. In an identical manner , the inductance of every other transmission line within the specimen shall also be measured. The characteristic impedance of each transmission line shall then be calculated by using the following formula:

$$\text{Characteristic impedance, in ohms} = (10^5 L/C_m)^{1/2}$$

Where: C_m = Mutual capacitance, in picofarads per foot, as calculated in 4.8.2.3
L = Measured inductance, in microhenries/10 feet.

(b) Specimens incorporating any number of individually shielded or unshielded insulated conductor pair transmission lines shall be laid straight. At one specimen end the two exposed insulated conductors of each transmission line shall be laid adjacent to each other (to minimize superfluous inductance) and electrically connected by tightly twisting both together. At the opposite specimen end, one transmission line shall be selected for measurement, and its two conductors connected electrically to the inductance measuring instrument. Its shield (if any) shall remain disconnected. The inductance of this transmission line

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shall be measured. In an identical manner , the inductance of every other transmission line within the specimen shall also be measured. The characteristic impedance of each insulated conductor pair shall then be calculated by using the formula shown in (a).

4.8.3.4 Observation. Specimen failure shall be construed if the specimen exhibits a characteristic impedance other than that specified (see 3.1).

4.8.4 Conductor resistance. This test shall determine whether or not conductors within the completed cable or cord are fabricated from the specified material.

4.8.4.1 Procedure. Conductor resistance shall be tested in accordance with ASTM B 193 and shall be corrected to 25°C.

4.8.4.2 Observation. Specimen failure shall be construed if any specimen conductor exhibits a conductor resistance greater than that specified (see 3.1).

4.8.5 Insulation resistance. This test shall determine the ability of completed cable to transmit d.c. without exhibiting unacceptable current flow through insulation of jacketing material, either between conductors, or between these conductors and any conducting substance which may be in contact with the surface of the cable or cord during in service use. The voltage with stand test (see 4.8.8) shall be performed prior to the insulation resistance test.

4.8.5.1 Specimen. The specimen shall be that which was previously used to perform the voltage withstand test (see 4.8.8.1).

4.8.5.2 Special apparatus. Apparatus shall include the following:

(a) A megohmmeter, megohm bridge or other suitable instrument which shall use a d.c. test voltage of not less than 200 V nor greater than 500 V, which shall support the requirements as specified in 4.8.5.3, and which shall make measurements within an accuracy of plus or minus 10 percent.
(b) Water tank (may not be required, see 4.8.5.3(d)) which shall be of size to contain the specimen, and which shall be suitable for the use as specified in 4.8.5.3(d).

4.8.5.3 Procedure. The specimen shall be maintained at a constant temperature, within plus or minus 3°C, for a period of not less than 5 hours immediately prior to testing. This temperature shall be noted and shall be used to calculate F (see formula below) when measurements are made in accordance with (a), (b) or (c). The megohmmeter shall then be electrically connected to specimen as specified (see (a), (b), (c) and (d)). For each connection, the test voltage from the megohmmeter shall be continuously applied to the specimen for a period of 1 minute, plus 25 percent, minus 0 percent, immediately following which the resistance reading of the megohmmeter shall be noted. Care shall be taken during each such measurement to assure that any electrical jumper leads connecting the specimen to the megohmmeter shall be spaced sufficiently

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far from each other and from any common conducting surface that measurement accuracy shall not be impaired.

(a) Specimens incorporating one or more individually shielded insulated conductor transmission lines: one transmission line shall be selected for measurement. One conductor of this transmission line shall be electrically connected to one terminal of the megohmmeter, and the shield and all other conductors (if any) of this transmission line shall be electrically connected to the other megohmmeter terminal. The resistance measurement shall then be performed as specified. In an identical manner, the resistance between each remaining conductor in this transmission line (if any) and the short circuited combination of all other conductors (if any), plus the shield in this transmission line shall be measured. Following this, and in an identical manner, resistance measurements shall be made for all remaining transmission lines within the specimen. (Additional measurements may be required, see (c) and (d).)

(b) Specimens incorporating multiple insulated conductors without individual shields: one insulated conductor shall be selected and electrically connected to one terminal of the megohmmeter. All other specimen conductors, including the overall shield (if any) shall be electrically connected to the other megohmmeter terminal. The resistance measurement shall then be performed as specified. In a similar manner, the resistance between each remaining conductor within the specimen and the short circuited combination of all other conductors plus the overall shield (if any) shall be measured. If the specimen incorporates an overall shield, then an additional resistance measurement shall be made between the shield and the short circuited combination of all remaining conductors. If the specimen does not incorporate an overall shield or armor over the jacket the test shall be made after placing the reel of cable in water for a period of not less than 1 hour. (Additional measurements may be required, see (d).)

(c) Specimens incorporating one or more individually shielded insulated conductor transmission lines (these measurements shall be required only when shield to shield measurements are specified, see 3.1): one shield shall be selected and electrically connected to one terminal of the megohmmeter. Other specimen shields shall be electrically connected to the other megohmmeter terminal, and the resistance measurement shall then be performed as specified. In a similar manner, the resistance between each remaining shield within the specimen and the short circuited combination of all other shields shall be measured.

(d) Specimens incorporating single or multiple insulated conductors without shield or armor and other specimens requiring measurements to water (see 3.1): the specimens shall be submerged for not less than 1 hour, except for its extreme ends (which shall remain exposed to air), in the water tank. The single conductor, or on multi conductor cables, the conductors are to be electrically connected to one terminal of the megohmmeter. For other specimens, the overall outer most shield (if any) or if there is no overall shield, the

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individual shields that are outermost in the specimen core shall be electrically connected to one terminal of the megohmmeter. The other megohmmeter terminal shall be electrically connected to the water surrounding the specimen (as by means of a bare metal wire which is both connected to the megohmmeter terminal and suspended within the water surrounding the specimen). The resistance measurement shall then be performed as specified. The temperature of the water surrounding the specimen shall also be measured and shall be used to determine F in the following formula:

The lowest value of resistance measured in accordance with each of (a), (b), (c) and (d), as applicable, shall be modified to include the effects of specimen length and temperature by using the following formula:

Insulation resistance, in megohms 1000 feet = RLF/1000

Where: R = Lowest value of resistance measured in accordance with (a), (b), (c) or (d), as applicable, in megohms.
L = Measured specimen length, in feet; measurement accuracy shall be within plus or minus 2 percent.
F = An ad hoc number, which shall accurately modify the measured resistance R to that value of resistance which would be measured if the specimen were at a uniform temperature of 15.5°C. This number shall be determined by the specimen manufacturer via ad hoc testing for the specific insulation or jacketing material separating the conductors between which the resistance measurement was made, and shall be normalized to unity for a material temperature of 15.5°C.

4.8.5.4 Observation. Specimen failure shall be construed if any calculated value of insulation resistance is less than that specified (see 3.1).

4.8.6 Mutual inductance. This test shall determine the ability of shielded, insulated conductor pairs within completed cable incorporating both shielded, insulated conductor pairs and unshielded insulated conductors to minimize crosstalk caused by the inductive effects of signal currents flowing in adjacent insulated conductors.

4.8.6.1 Specimen. The specimen shall be that which was previously used to determine characteristics impedance (see 4.8.3.1).

4.8.6.2 Special apparatus. Apparatus shall include the following:

(a) A sinusoidal signal generator, which shall produce an unchanging signal at a constant frequency of 1000 ±200 Hz, which shall produce a signal current appropriate for use in the test as specified in 4.8.6.3 (a signal current of approximately 1 A is typically found necessary), and the signal frequency and signal current shall be known within an accuracy of plus or minus 2 percent.

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(b) A rms indicating microvoltmeter, intended for use at the signal generator frequency, exhibiting a signal input impedance not less than 10,000 ohms, shall be used for the specimen measurements as specified in 4.8.6.3, and shall exhibit a measurement error of not more than plus or minus 6 percent.

4.8.6.3 Procedure. One shielded, insulated conductor pair within the specimen shall be selected, and the microvoltmeter shall be electrically connected across this conductor pair at its unshorted end. The shield shall remain disconnected. The signal generator shall then be electrically connected across any two unshielded, insulated conductors at the specimen end opposite that of the microvoltmeter connection. These same two unshielded, insulated conductors shall be connected together electrically at the opposite specimen end (the end with the attached microvoltmeter) by twisting their two conductors tightly together. The signal current from the signal generator shall next be adjusted until a reasonable indication is noted on the microvoltmeter, and both the microvoltmeter reading and the signal current magnitude shall then be recorded. In a similar manner, additional measurements shall be taken such that microvoltmeter measurements are taken for each shielded, insulated conductor pair as the signal generator is sequentially reconnected to circulate signal current in each possible pairing of unshielded, insulated conductors. The mutual inductance represented by each measurement shall then be calculated by using the following formula:

$$\text{Mutual inductance, in microhenries} = (V/2)\ 3.14159\ fI$$

Where: V = Measured potential between two shielded, insulated conductors, in microvolts.
f = Signal frequency, in Hz.
I = Measured signal current, in A.

4.8.6.4 Observation. Specimen failure shall be construed if any calculated value of mutual inductance is greater than that specified (see 3.1).

4.8.7 Pulse response time. This test shall determine the ability of completed cable cord which incorporates one or more shielded, insulated conductor pairs for pulse data transmission to propagate a square wave signal without causing unacceptable signal distortion.

4.8.7.1 Specimen. The specimen shall consist of a piece of completed cable, which shall have a length of not less than 200 feet. Individual transmission lines within this specimen (where each insulated conductor pair plus surrounding shield is considered to be a transmission line) shall be selected for testing as follows:

(a) Specimens containing four or fewer transmission lines: every transmission line shall be tested.

* (b) Specimens containing more than four but fewer than 21 transmission lines: Four different transmission lines shall be tested. Not fewer than one of these transmission lines shall be selected from each concentric cabling layer within the specimen.

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(c) Specimens containing 21 or more transmission lines: the number of transmission lines tested shall be equal to the square root (rounded, if necessary, to the nearest whole number) of the total number of transmission lines. Not fewer than one of these transmission lines shall be selected from each concentric cabling layer within the specimen.

4.8.7.2 Special apparatus. Apparatus shall include the following:

(a) Pulse generator: a signal generator which shall produce uniform square wave voltage pulses at a constant rate of repetition, and which shall be used for the specimen measurements as specified in 4.8.7.3 The pulses produced by this generator shall each exhibit a 10 to 90 percent voltage rise time of not more than 50 nanoseconds, a 90 to 10 percent voltage decay time of not more than 50 nanoseconds, and a pulse width of not less than 2 microseconds.

(b) Oscilloscope: a dual beam oscilloscope shall have a band width of not less than 10 megahertz, and a time base capable of producing a horizontal beam sweep rate of not slower than 0.1 milliseconds (ms) per centimeter. This oscilloscope shall have provision for a high impedance, balanced or differential input on both signal channels, and shall be used for the specimen measurements as specified in 4.8.7.3.

(c) Pulse transformers: two, winding transformers, each of which shall provide essentially distortion free transmission of the voltage pulses from the pulse generator, and shall be used for the specimen measurements as specified in 4.8.7.3 One of the pulse transformers (the "input" pulse transformer, see figure 18) shall have a primary winding impedance equal to the nominal signal terminal impedance of the pulse generator; the secondary winding of this transformer shall be center tapped, and shall have an impedance equal to the specified impedance of the specimen (see 3.1). The other pulse transformer (the "output" pulse transformer, see figure 18) shall have a primary winding which shall be center tapped, and shall have an impedance equal to the specified impedance of the specimen (see 3.1); the secondary winding impedance of this transformer shall equal the nominal resistance of the matching resistor (see (d)).

(d) Matching resistor: an electrical resistor, shall be essentially noninductive in construction (such as a carbon type resistor), which shall have a power rating appropriate for its intended use (see 4.8.7.3), and shall exhibit an electrical resistance appropriate for use with the output pulse transformer (see (c)).

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4.8.7.3 Procedure. The test apparatus shall be electrically interconnected as shown on figure 18, one of the selected transmission lines shall be connected a shown. The connections between this transmission line and the pulse transformers shall be as short as practicable. In addition, both connecting cables between the transmission line and the oscilloscope shall be of identical length and construction and shall not be longer than 5 feet each in order to minimize pulse distortion. the pulse generator shall then be turned on, and both signal channels on the oscilloscope shall be adjusted such that the pulse waveform displayed by each on the oscilloscope screen shall be of the same height. The horizontal beam sweep rate of the oscilloscope shall then be increased, if necessary and the time required for the leading edge of each signal channel pulse to rise from 10 to 90 percent of its final value shall be measured as shown on figure 19. The rise time of the transmission line shall then be calculated by using the following formula:

$$\text{Rise time, in ns per 500 feet} = 500(Y_r^2 - X_r^2)^{1/2}/L$$

Where: X_r = Measured rise time from signal channel number one, in microseconds
Y_r = Measured rise time from signal channel number two, in microseconds.
L = Measured specimen length, in feet; measurement accuracy shall be within plus or minus 1 percent.

Following this, the time required for the trailing edge of each signal channel pulse to decay from 90 to 10 percent of its initial value shall be measured as shown on figure 20. The decay time of the transmission line shall then be calculated by using the following formula:

$$\text{Decay time, in ns per 500 feet} = 500(Y_d^2 - X_d^2)^{1/2}/L$$

Where: X_d = Measured decay time from signal channel number one, in microseconds.
Y_d = Measured decay time from signal channel number two, in microseconds.
L = Measured specimen length, in feet; measurement accuracy shall be within plus or minus 1 percent.

In an identical manner, the remaining selected transmission lines shall also be tested.

4.8.7.4 Observation. Specimen failure shall be construed if any transmission line exhibits either a rise time or a decay time which is greater than that specified (see 3.1).

4.8.8 Voltage withstand. This test shall determine the ability of completed cable to withstand overvoltages without exhibiting electrical discharge through insulation or jacketing material, either between conductors, or between these conductors and any conducting material which may be in contact with the surface of the cable during in service use.

* 4.8.8.1 Specimen. The specimen shall consist of a unit of product as specified in 6.7.1.1.

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4.8.8.2 Special apparatus. Apparatus shall include the following:

(a) An adjustable voltage source, which shall be in accordance with the following.

(1) An a.c. voltage source which shall produce a single phase, approximately sinusoidal voltage under all test conditions, which shall have a frequency of not more than 100 Hz, and which can be adjusted to the specified rms magnitude (see 3.1). This source shall have the capacity to provide an effective power of not less than 5 kVA, and shall incorporate an overload relay which shall reduce the source voltage to zero whenever an applied load (that is, the specimen, see 4.8.8.3) attempts to draw electrical current to exceed this effective power capacity.

(b) When required, a water tank, which shall be filled with water, which shall be of sufficient size to contain the specimen, and which shall be used as specified in 4.8.8.3(d) (may not be required, see 4.8.8.3(d)).

* 4.8.8.3 Procedure. The voltage source shall be electrically connected to the specimen as specified (see (a), (b), (c) and (d)). For each connection, the test voltage from the voltage source shall be increased, in a uniform manner, from o v to the specified voltage (see 3.1) during a period of not more than 60 seconds nor less than 10 seconds, except that specified voltages of less than 600 v may be applied instantaneously. (This gradual voltage application is generally necessary to prevent a current on rush, and the consequent tripping of the voltage source overload relay caused by specimen capacitance.) This specified voltage shall then be maintained for a period of 1 plus 1/2, minus o minute (unless the voltage source overload relay trips, see 4.8.8.4), following which the test voltage shall be reduced to zero at approximately the same rate as which it was applied.

(a) Specimens incorporating one or more individually shielded, insulated conductor transmission lines: one transmission line shall be selected for measurement. One conductor of this transmission line shall be electrically connected to one terminal of the voltage source, and the shield and all other conductor(s) (if any) of this transmission line shall be electrically connected to the other voltage source terminal. The voltage withstand test shall then be performed as specified. In a similar manner, the voltage withstand test shall be performed between each remaining conductor in this transmission line (if any) and the short circuited combination of all other conductors (if any) plus the shield in this transmission line. Following this, and in an identical manner, the voltage withstand test shall be performed on all remaining transmission lines with the specimen. (Additional measurements may be required, see (c) and (d).)

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(b) Specimens incorporating multiple insulated conductors without individual shields: one insulated conductor shall be selected and electrically connected to one terminal of the voltage source. Other specimen conductors, including the overall shield (if any) shall be electrically connected to the other voltage source terminal. The voltage withstand test shall then be performed as specified. In a similar manner, the voltage withstand test shall be performed between each remaining conductor within the specimen and the short circuited combination of all other conductors plus the overall shield (if any). If the specimen incorporates an overall shield, then the voltage withstand test shall additionally be performed between the shield and the short circuited combination of all remaining conductors. (Additional measurements may be required, see (d).)

(c) Specimens incorporating one or more individually shielded insulated conductor transmission lines (these measurements shall be required only when shield to shield measurements are specified, see 3.1): one shield shall be selected and electrically connected to one terminal of the voltage source. other specimen shields shall be electrically connected to the other voltage source terminal, and the voltage withstand test shall then be performed as specified. In a similar manner, the voltage withstand test shall be performed between each remaining shield within the specimen and the short circuited combination of all other shields.

(d) All specimens (these measurements shall be required only when measurements to water or to ground are specified, see 3.1): the specimen shall be submerged, except for its extreme ends (which shall remain exposed to the air), in the water tank for a period of not less than 1 hour. Following this, and while the specimen is still submerged, the specimen shall be tested as follows. The overall specimen shield (if any), or if there is no overall shield, all individual shields which are outermost in the specimen core plus all unshielded conductors which are outermost in the specimen core, shall be electrically connected to one terminal of the voltage source. The other voltage source terminal shall be electrically connected to the water surrounding the specimen (as by means of a bare metal wire which is both connected to the voltage source terminal and suspended within the water surrounding the specimen). The voltage withstand test shall then be performed as specified.

(e) Specimens incorporating single or multiple insulated conductors without shield or armor shall be tested to water. The specimen shall be submerged, except for its extreme ends (which shall remain exposed to the air), in the water tank for a period of not less than 1 hour. While the specimen is still submerged, the specimen shall be tested as follows. For single conductors, the conductor shall be electrically connected to one terminal of the voltage source. The other voltage source terminal shall be electrically connected to the water

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surrounding the specimen (as by means of a bare metal wire which is both connected to the voltage source terminal and suspended within the water surrounding the specimen). For multiple conductor cables, one insulated conductor shall be selected and electrically connected to one terminal of the voltage source. Other specimen conductors, shall be electrically connected to the other voltage source terminal. The voltage withstand test shall then be performed sequentially on each conductor as specified.

4.8.8.4 Observation. Specimen failure shall be construed if the voltage source overload relay trips prior to removal of the test voltage from any specimen connection. If the insulation resistance test is specified (see 3.1), then the specimen shall be retained for use in that test.

4.8.9 Jacket flaws. One hundred percent of all finished cables with overall shields shall be passed through a chain electrode spark test device using 3.0 kV rms (minimum) and a frequency of 50 or 60 Hz. The conductor, or shield as applicable, shall be grounded at one or both ends. The electrode shall be of a suitable bead chain or fine mesh construction that will give intimate metallic contact with practically all the insulation surface. Electrode length and speed of specimen movement shall be such that the insulation is subjected to the test voltage for a minimum of 0.2 second.

4.8.9.1 Observation. Specimen failure shall be construed if jacket material exhibits any sign of deformation or damage.

4.8.10 Conductor and shield continuity. One hundred percent of all finished cable shall be tested for continuity prior to shipment. To establish continuity, not more than 25 volts d.c. shall be applied across each conductor and shield of the cable through an appropriate indicator, such as an ohmmeter, light or buzzer. The test voltage may be applied to conductors and shields individually or in series.

4.8.10.1 Observation. Specimen failure shall be construed if the specimen exhibits an open electrical circuit.

* 4.8.11 Surface transfer impedance. The surface transfer impedance characteristics of the overall cable shield of finished cable shall be tested in accordance with NEMA WC-61, and as specified in 4.8.11.1. The electromagnetic pulse (EMP) response of the cable shield shall be determined in accordance with 4.8.11.2.

* 4.8.11.1 Cable sample preparation. The individual shields of shielded components, when present, shall be connected to the conductors and the connected conductors and shields shall be the conductor as specified in NEMA WC-61.

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* 4.8.11.2 Determination of EMP response. The surface transfer impedance determined as specified shall be known over the frequency range d.c. to 100 MHz. Z_t shall be assumed constant in the frequency range d.c. to 0.1 MHz. The EMP response in dB shall be determined as follows:

$$\text{EMP response} = -185 - 10 \log \left[\int_{0}^{1 \times 10^8} \frac{[Z_t(f)]^2 df}{(\hat{A}^2 + f^2)(\beta^2 + f^2)} \right] \text{ (dB)}$$

Where: $Z_t(f)$ = surface transfer impedance at frequency f (õ/m)
$\hat{A} = 2.39 \times 10^5$
$\beta = 4.12 \times 10^7$
f = frequency (Hz)
df = decade of frequency

The integral expression may be approximated by a summation if at least 100 points per decade of frequency are used.

4.8.11.3 Observation. Specimen failure shall be construed if the surface transfer impedance is greater than specified (see 3.1) or if the calculated EMP response of the overall cable shield is less than specified (see 3.1).

4.9 Inspection of packaging. Sample packages and packs, and the inspection of the preservation packaging, packing and marking for shipment and storage shall be in accordance with the requirements of section 5 and the documents specified therein.

5. PACKAGING

(The preparation for delivery requirements specified herein apply only for direct Government acquisition. For the extent of applicability of the preparation for delivery requirements of referenced documents listed in section 2, see 6.5.)

5.1 Shipment of cables. Cables covered by this specification shall be shipped on reels, unless otherwise specified on the specification sheet.

5.1.1 Reels. Reels shall be of a design, strongly constructed, with flanges of sufficient diameter to protect the cable from injury during handling and storage. Where the gross weight of cable and reel exceed 1000 pounds, the reels shall have metal hubs. The diameter of the drum shall be not less than 14 times the overall diameter of the cable.

5.1.1.1 Placement of cable on the reel. The cable on each reel shall be one continuous length and shall have both ends readily available for testing without re-reeling. The inner end shall extend through a hole in the flange near the drum, and the outer end shall be available at the top of the reel. Both ends of the cable shall be secured to reel flange with 1/4-inch rope, applied with two half-hitches, one 6 inches and the other 4 inches (approximately) back of the end seal, then twisting the rope ends together, knotting them, pulling the cable up tight and stapling the rope to the flange with at least two 1-1/2 inch staples driven into the flange. The inner end which extends through the flange hole shall be protected by a metal end cover.

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5.1.1.2 Lagging. Lagging shall be used on reels having flange diameters of 42 inches and larger and on all reels which contain 1000 pounds or more of cable. Reels smaller than 42 inches in diameter which contain less than 1000 pounds of cable may be covered in accordance with 5.1.1.3. Lagging boards shall have a thickness equal to the thickness of the reel flange but not greater than 2 inch commercial lumber. Lagging for all reels shall extend to within 1/4 inch of the outside of the flanges. Lagging boards shall be installed so that they are touching adjacent boards. Lagging which rest entirely on the rim of the flange shall be nailed with cement-coated nails. Lagging which is notched and recessed may be secured by the use of stapling only. In addition to nailed lagging or for notched and recessed lagging on reels of cable weighing 1000 pounds or more and having a flange diameter less than 78 inches, two 0.162 ± 0.003 inch (8-gage) high-tensile round steel wires or two 3/4 by 0.035 inch steel straps shall be used. For reels having a flange diameter of 78 inches or more two 1-1/4 by 0.035 inch steel straps shall be used. In addition to nailed lagging or for notched and recessed lagging on reels of cable weighing less than 1000 pounds, two 0.0915 ± 0.003 inch (13-gage) high-tensile round steel wires or two 5/8 by 0.020-inch steel straps shall be used. Straps shall be stapled at approximately 15-inch intervals. A minimum clearance of 2 inches shall be provided between inner face of lagging and outer layer of cable.

5.1.1.3 Alternative to lagging. As an alternative to lagging for reels having a flange diameter smaller than 42 inches and containing less than 1000 pounds of cable, a covering may be used which consists of the following:

(a) A cushioning inner layer of asphalt-impregnated, corrugated, fiberboard, or the equivalent, wrapped to give 100-percent coverage.
(b) An outer covering consisting of two layers of weather resistant veneer paper in accordance with PPP-V-205 or two layers of three ply weather resistant toxic treated plywood wrapped to give 100-percent coverage.

This covering shall be secured in place between the reel flanges by at least two 3/8 by 0.020 inch steel straps.

* 5.1.1.4 Year marking. In addition to any other required marking, reels shall be marked with a keyed series of colors to indicate the year of manufacture. The cycle of colors shall be repeated every fifth year. The reel marking shall consist of a stripe of one coat of commercial quality outside paint approximately 2-inches wide, colored to designate the particular year of manufacture as follows:

Year of manufacture			Identifying color
1991	1996	2001	Red
1992	1997	2002	Green
1993	1998	2003	Orange
1994	1999	2004	Blue
1995	2000	2005	White

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5.1.1.4.1 Location of year marking. The colored stripe on reels shall be applied circumferentially over the lagging or the alternative to lagging and midway between the flanges. The stripe shall consist of one coat of commercial quality outside paint of the appropriate color. In addition to the stripe, both flanges of the reel shall be stenciled with 4 inch high figures to show the year of manufacture.

5.1.1.5 Standard reel markings. Each reel shall be plainly marked on both flanges with the following information:

(a) Reel number.
(b) Type and size of cable.
(c) Footage.
(d) Contract or order number.
(e) Contractor's name.
(f) Manufacturer's name (if other than contractor)
(g) Gross weight.
(h) One continuous length or the sequence of lengths on the reel indicated in feet, whichever is applicable.

5.1.2 Coils. Coils shall contain one continuous length of cable (standard length, see table XIII).

* TABLE XIII. Cable lengths.

				Remnant lengths Price reduction		
Type of package	Nominal length (feet)	Standard lengths No price reduction (feet)	Random lengths No price Reduction (feet)	5 percent (feet)	10 percent (feet)	Scrap lengths Not acceptable (feet)
Coil	500	550 to 450	449 to 150	149 to 100	99 to 50	49 to 0
Coil	600	660 to 540	539 to 180	179 to 120	119 to 60	59 to 0
Reel	250	300 to 200	199 to 100	99 to 75	74 to 50	49 to 0
Reel	500	550 to 450	449 to 150	149 to 100	99 to 50	49 to 0
Reel	800	880 to 720	719 to 240	239 to 160	159 to 80	79 to 0
Reel	1000	1100 to 900	899 to 300	299 to 200	199 to 100	99 to 0

5.1.2.1 Year marking. In addition to any other required marking, coils shall be marked with a keyed series of colors to indicate the year of manufacture. This marking shall consist of a stripe approximately 2-inches wide and colored for the particular year of manufacture. The cycle of colors shall be the same as those used for year marking of reels (see 5.1.1.4).

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5.1.2.2 Identification. Two shipping tags shall be securely attached to each coil, both inside and outside the wrapping, and marked with the following information.

(a) Type and size of cable.
(b) Footage.
(c) Contract or order number.
(d) Contractor's name.
(e) Manufacturer's name (if other than contractor).
(f) Gross weight,

5.2 Packaging and packing. Cable as specified (see 6.2.1) shall be packaged and packed level A, B, or C in accordance with MIL-C-12000, as specified (see 6.2.1). Cables shall have both ends of each shipping length sealed to prevent the entrance of moisture. Materials used and method of application shall be that specified for tape wrapped end seals in accordance with MIL-C-12000. Marking shall be in accordance with MIL-C-12000 and with the requirements specified herein or as specified in acquisition document (see 6.2.1). All markings shall conform to MIL-STD-129.

6. NOTES

6.1 Intended use. Cables specified herein are intended for use in various applications involving naval ships and shore stations.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

(a) Title, number and date of this specification.
(b) Military part number (see 1.2).
(c) Type and size of cable or cord required with reference to applicable specification sheet (see 3.1).
(d) Level of packaging and packing required (see 5.2).
(e) Special marking required (see 5.2)
(f) Quantity required (see 6.2.3).

6.2.2 Data requirements. When this specification is used in an acquisition which incorporates a DD Form 1423, Contract Data Requirements List (CDRL), the data requirements identified below shall be develcped as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved CDRL incorporated into the contract. When the provisions of FAR 52.227-7031 are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification is cited in the following paragraphs

Paragraph no.	Data requirement title	Applicable DID no.	Option
3.3.16	Certification/data report	UDI-A-23264	------
4.4.1	Inspection system program plan	DI-R-4803	------
4.4.7	Reports, test	DI-T-2072	10.1.b

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(Data item descriptions related to this specification, and identified in section 6 will be approved and listed as such in DoD 5000.19L., Vol.II, AMSDL. Copies of data item descriptions required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

6.2.2.1 The data requirements of 6.2.2 and any task in sections 3, 4, or 5 of this specification required to be performed to meet a data requirement may be waived by the contracting/acquisition activity upon certification by the offeror that identical data were submitted by the offeror and accepted by the Government under a previous contract for identical item acquired to this specification. This does not apply to specific data which may be required for each contract regardless of whether an identical item has been supplied previously (for example, test reports).

6.2.3 Quantity. The quantity of each type and size on a contract or order should be specified as an integral multiple of the unit ordering length shown by the specification sheet.

6.2.4 Lengths. The range of standard, random, remnant and scrap lengths for each nominal length is shown in table XIII. In order to compensate for handling and probable loss on issue, remnant lengths are subject to the price reductions specified in table XIII. Scrap lengths are not acceptable. It is not intended that items being acquired in exact lengths for a specific job or ship shall necessarily be shipped in lengths which are integral multiples of the unit ordering length or that price reductions shall apply in the event that exact footages or fractional lengths are required and so indicated in acquisition of cable other than stock purposes.

6.3 Special clause. Except when small quantities are to be purchased, invitations for bid and contracts or orders should specify the following:

"In order to compensate for handling and probable loss in issue, remnant lengths shall be subject to the price reductions specified in table XIII. In order that the number of lengths be kept to a minimum consistent with good manufacturing practices, for each type and size of cable or cord on the contract or order, not less than 70 percent of the total footage to be shipped shall be in standard lengths, and not more than 30 percent may be in any combination of random and remnant lengths. When the total quantity of any one item is six nominal lengths or less, a footage approximately two nominal lengths (in lieu of 30 percent) may be in any combination of random and remnant lengths. When the total quantity of any one item is two nominal lengths or less, the total footage may be in any combination of random lengths."

6.4 Time delay. Comparison inspection will normally require 40 calendar days from the date the sample is sent to the Government designated laboratory to the date the inspection results are received by the manufacturer. This time delay should be taken into consideration by a contractor when estimating delivery time.

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6.5 With respect to products requiring qualification, awards will be made only for products which are, at the time set for opening of bids, qualified for inclusion in Qualified Products List QPL-24640 whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or purchase orders for the products covered by this specification. The activity responsible for the Qualified Products List is the Naval Sea Systems Command, SEA 55Z3, Department of the Navy, Washington, DC 20362 and information pertaining to qualification of products may be obtained from that activity. Application for Qualification tests shall be made in accordance with "Provisions Governing Qualification SD-6" (see 6.5.1).

6.5.1 Copies of "Provisions Governing Qualification SD-6" may be obtained upon application to Commanding Officer, Naval Publications and Forms Center, 5801 Tabor Avenue, Philadelphia, PA. 19120.

6.6 Sub-contracted material and parts. The preparation for delivery requirements of referenced documents listed in section 2 do not apply when material and parts are acquired by the contractor for incorporation into the equipment and lose their separate identity when the equipment is shipped.

6.7 Terms and definitions. For the purpose of the quality conformance inspection, the sampling terms specified in 6.7.1 through 6.7.4 shall apply.

* 6.7.1 Unit of product. A unit of product is the unit ordering length as specified on the specification sheet, except that when the unit ordering length is 250 feet, two such lengths shall constitute a unit of product. Random and remnant lengths or special order lengths may be added together to equal a unit ordering length. Length tolerances as given for standard lengths shall apply (see 6.2.3).

* 6.7.1.1 For electrical test purposes a unit of product shall be defined as an electrically continuous splice free length. This will allow for longer production lengths to be tested electrically prior to preparation for shipment.

* 6.7.1.2 Lengths longer than the unit ordering lengths (see 3.1) may be supplied when the end user and manufacturer agree. These lengths shall be as specified in 6.7.1.1

6.7.2 Sample. A sample is one unit of product selected from an inspection lot.

6.7.3 Specimen. A specimen is an individual length of cable or an individual length of a part of cable which has been taken from a sample.

6.7.4 Defective unit. A defective unit is a unit of product from which a specimen that failed to meet one or more requirements was taken.

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6.8 Part number or identifying number (PIN). The part number shall be a combination of numbers and letter in accordance with the following:

M	24640	/XX	-XX	A or U	O or D or N
Military designation	Specification Number	Specification Sheet number	Dash number (See Specification sheets)	A=Armored U=Unarmored	O=Overall D=Double overall shielded N=Not shielded

* 6.9 Subject term (Keyword) listing.

a. Armor
b. Waterblock
c. Halogen content
d. Smoke Index
e. Flame Propagation
f. Toxicity Index

NOTE: The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:
Army - MI
Navy - SH

Review Activities:
Army - AV, CR, ER, ME, AR, AL
Navy - EC, CG
DLA - IS

Preparing Activity:
Navy - SH

Agent:
DLA-IS

(Project No. 6145-2105)

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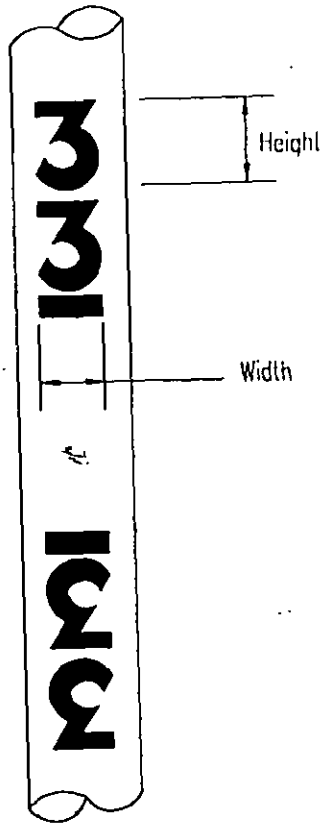


FIGURE 1. Conductor of pair number 33.

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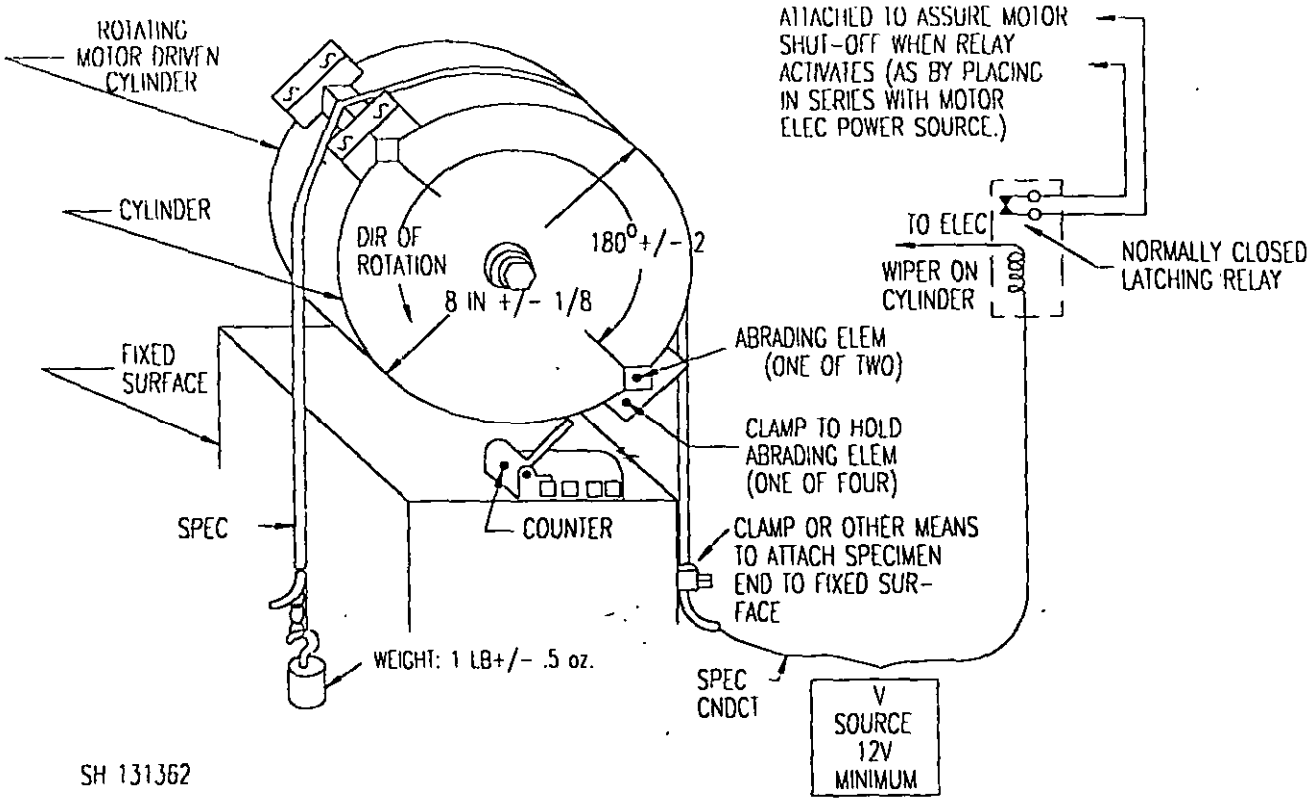


FIGURE 2. Diagram of abrading machine for the abrasion resistance test

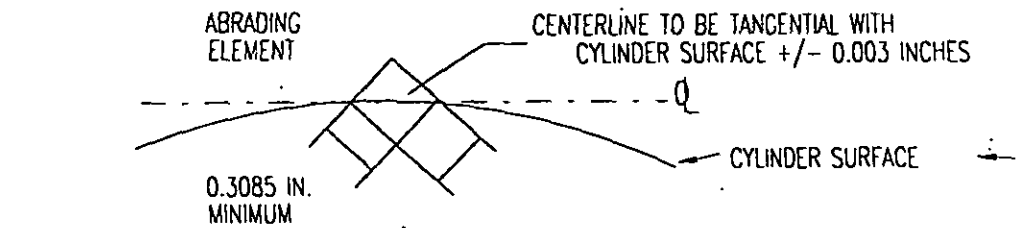
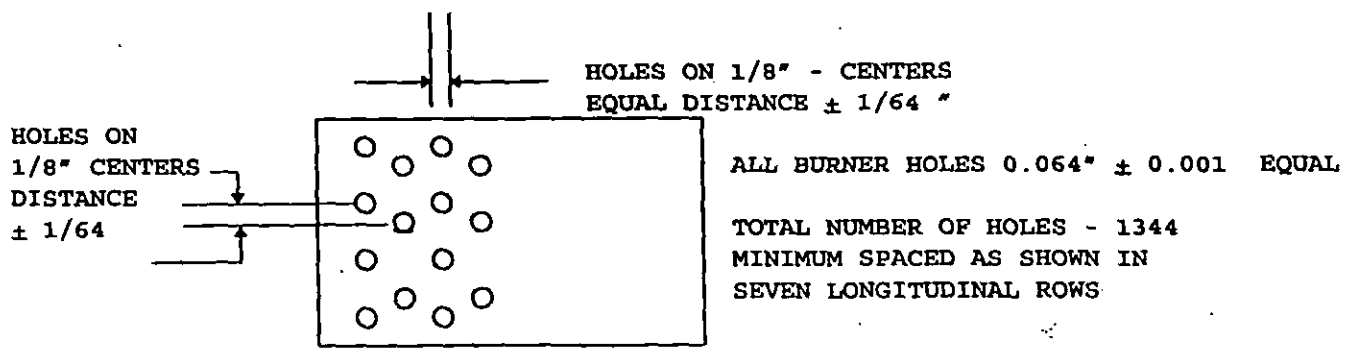


FIGURE 3. Detail diagram of abrading element fit on cylinder surface.



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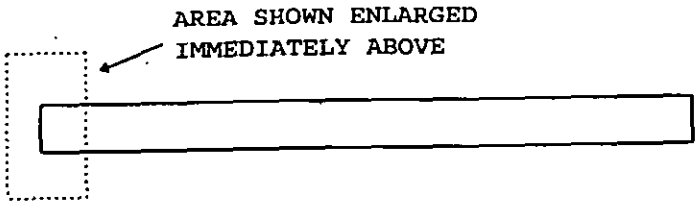


FIGURE 4. Top view of burner for gas flame test.

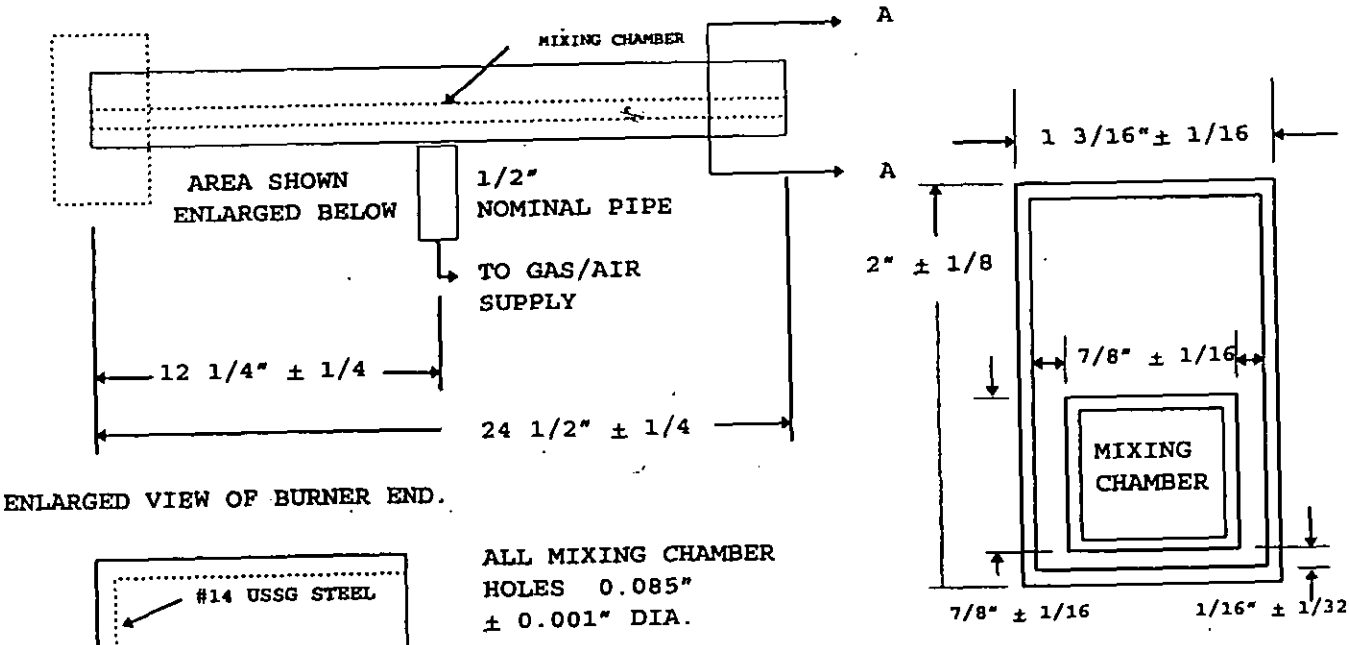


FIGURE 4(A). Side view of burner showing internal mixing chamber for the gas flame test

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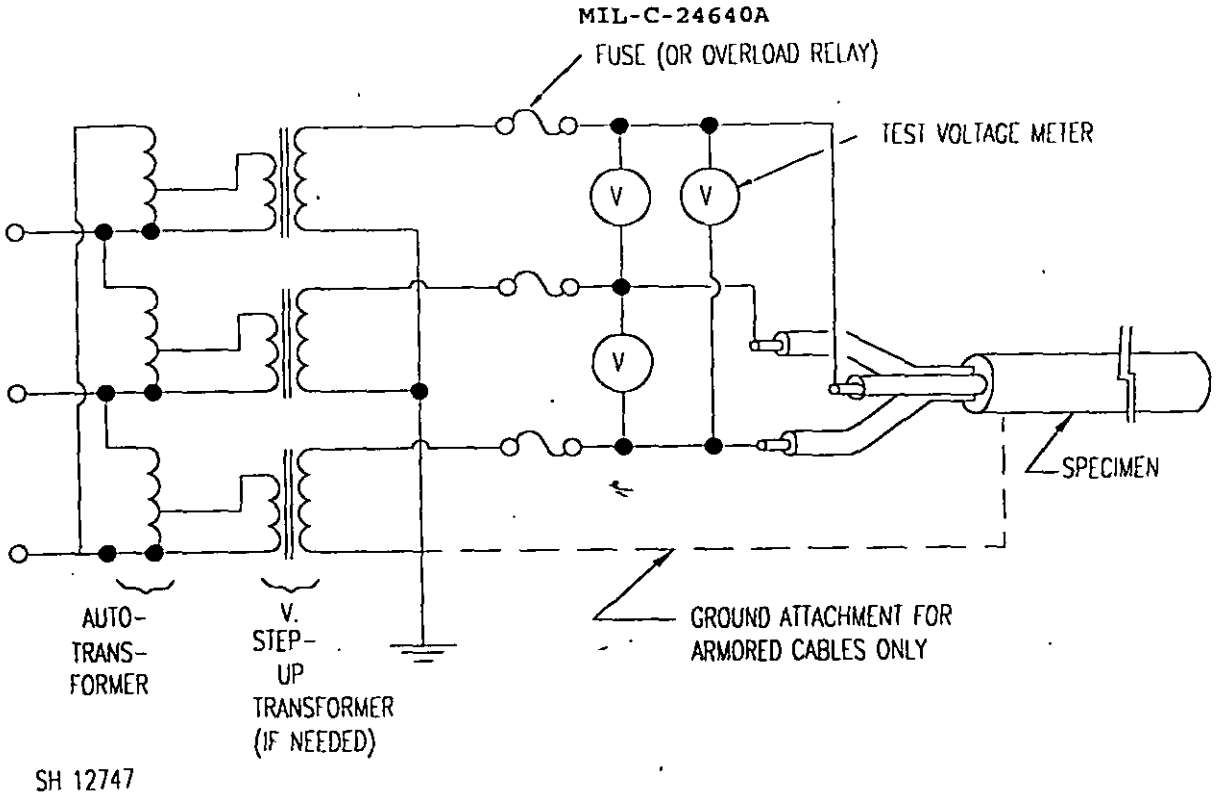
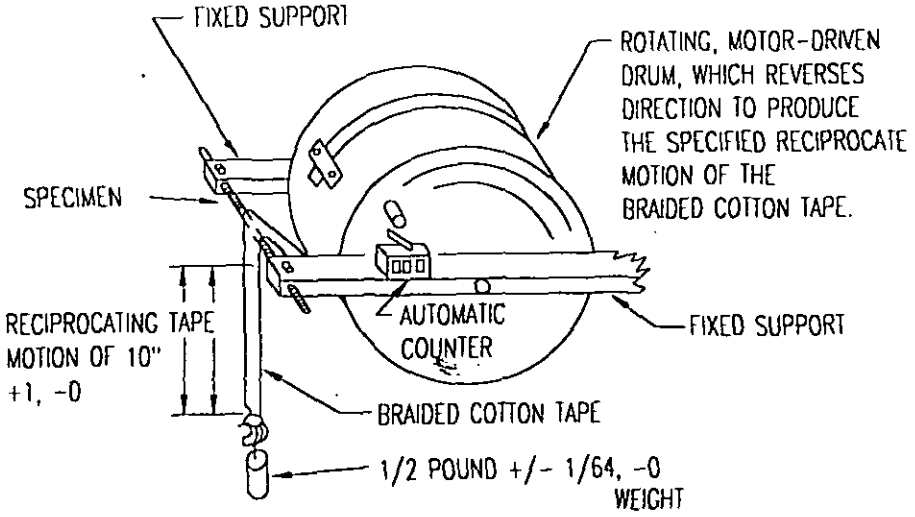


FIGURE 5. Application of test voltages for the gas flame test.

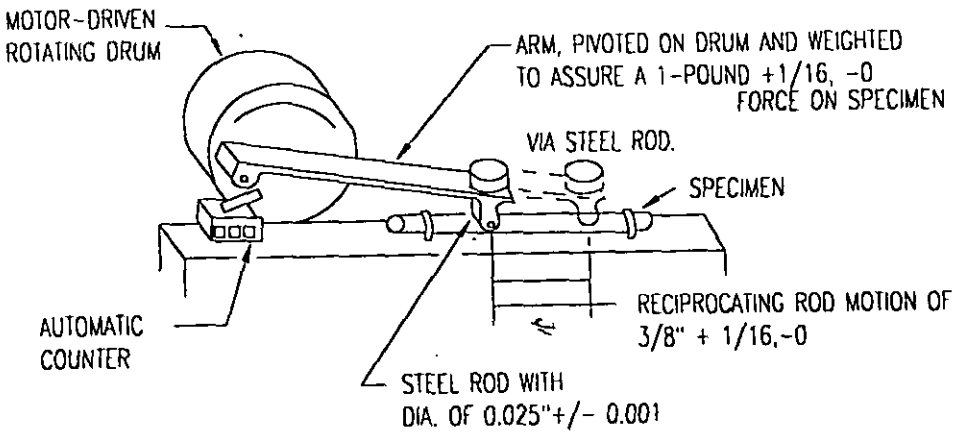
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SH 13062

Figure 6. Diagram of typical abrading machine for the permanence of printing (conductor insulation) test.

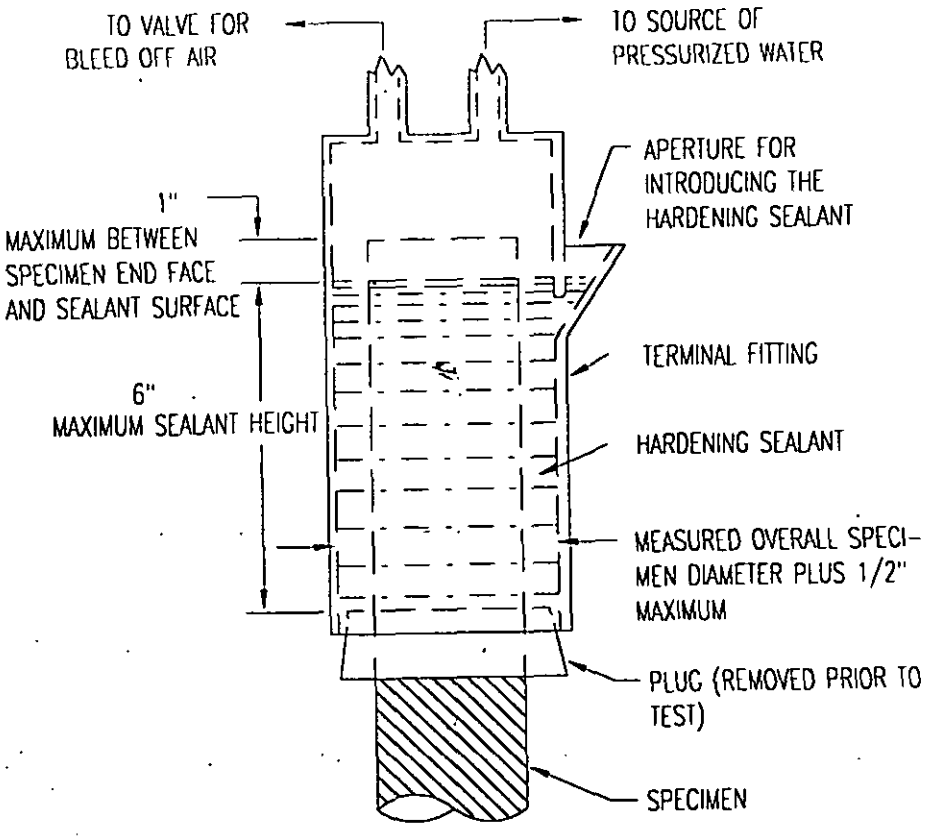
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SH 13063

FIGURE 7. Diagram of abrading machine for the permanence of printing (jacket) test.

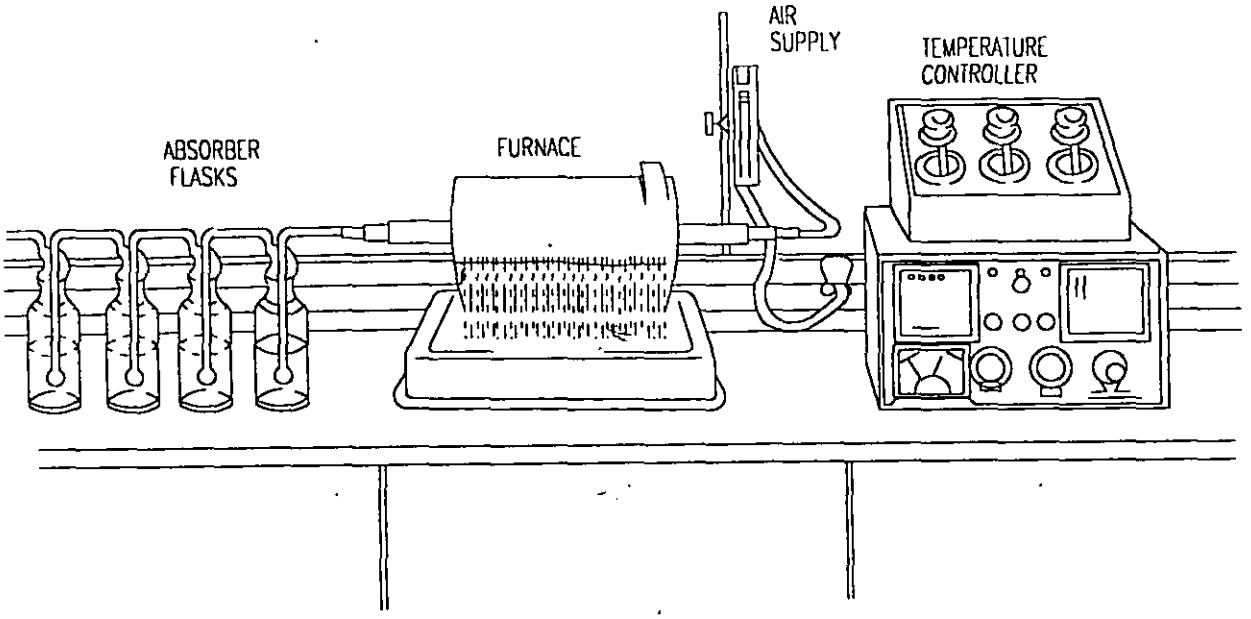
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SH 13064

FIGURE 8. Diagram of terminal fitting, showing internal details, for the watertightness test.

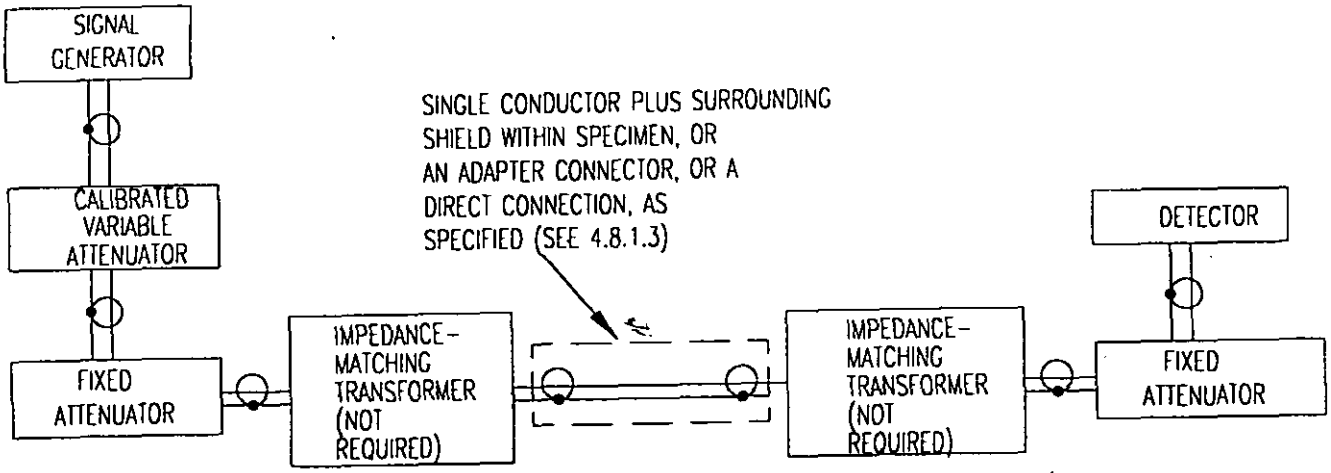
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SH 12742

FIGURE 9. Acid gas test apparatus.

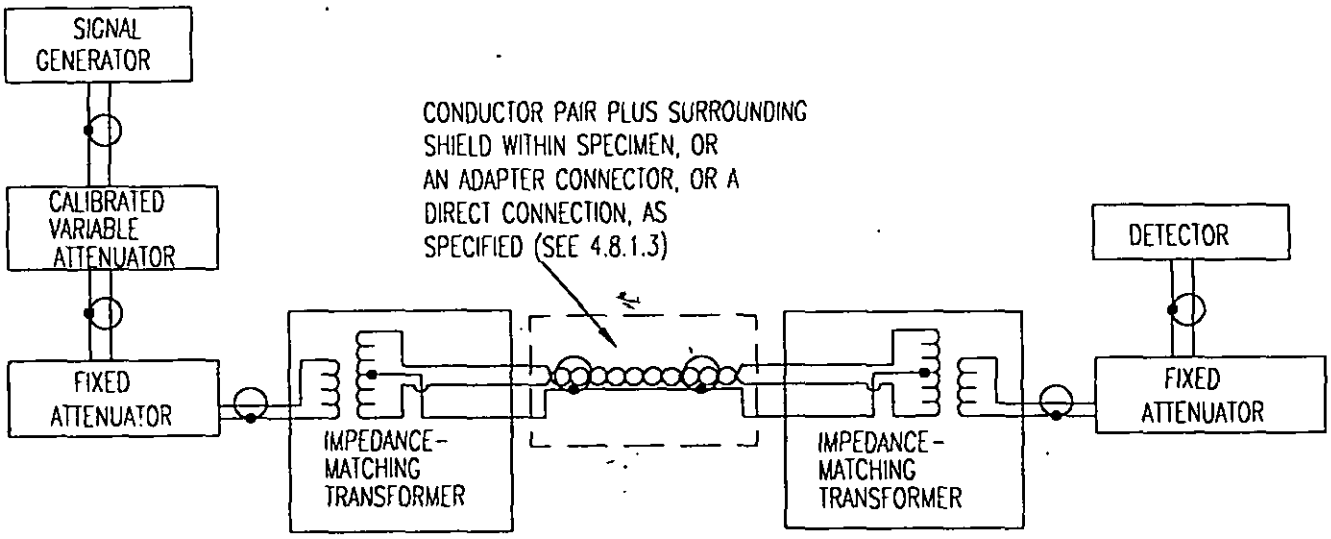
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SH 13065

FIGURE 10. Block diagram of electrical connections for the attenuation test when testing a single conductor plus surrounding shield.

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FIGURE 11. Block diagram of electrical connections for the attenuation test when testing a conductor pair plus surrounding shield.

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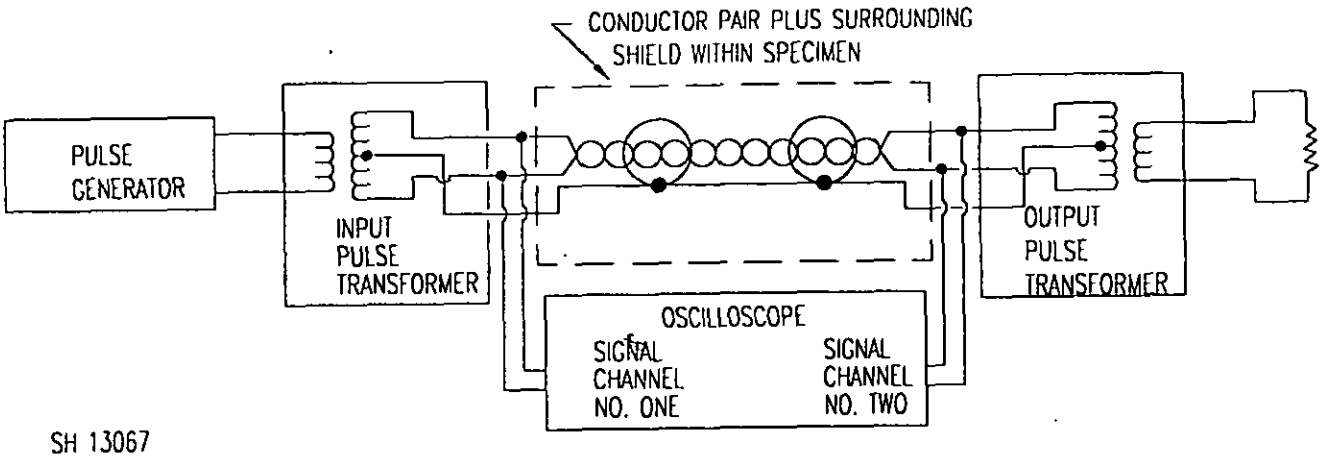


FIGURE 12. Block diagram of electrical connections for the pulse response test.

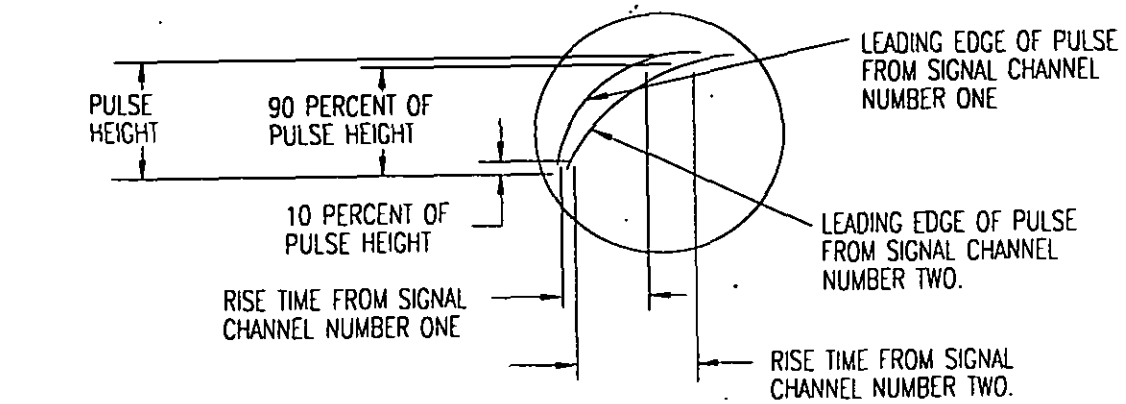
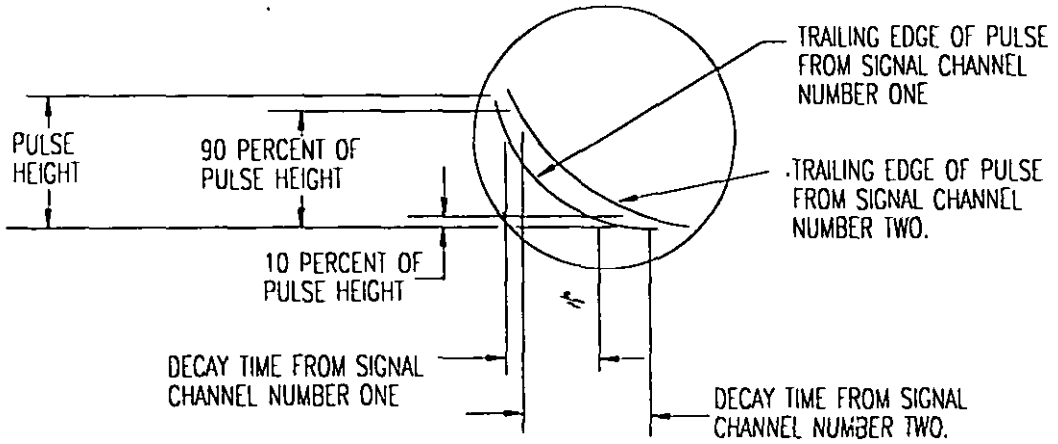


FIGURE 13. Oscilloscope display showing rise time measurements for the pulse response time test.

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SH 13069

FIGURE 14. Oscilloscope display showing decay time measurements for the pulse response time test.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-C-24640A	2. DOCUMENT DATE (YYMMDD) 950626

3. DOCUMENT TITLE: CABLES, LIGHT WEIGHT, ELECTRIC, LOW SMOKE, FOR SHIPBOARD USE, GENERAL SPECIFICATION FOR

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach sheets if needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER		
a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY DLA-IS

a. NAME Emelia Altomari	b. TELEPHONE (Include Area Code) (1) Commercial (215) 697-6827
c. ADDRESS (Include Zip Code) Defense Industrial Supply Center 700 Robbins Avenue Bldg. 3 (Code DISC-EED) Philadelphia, PA 19111-5096	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403 Falls Church, VA 22041 TELEPHONE (703) 756-2340 AUTOVON 289-2340

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